

RESEARCH ARTICLE

An extended minimal model of OGTT: estimation of α - and β -cell dysfunction, insulin resistance, and the incretin effect

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Abstract

Loss of insulin sensitivity, α - and β -cell dysfunction, and impairment in incretin effect have all been implicated in the pathophysiology of type 2 diabetes (T2D). Parsimonious mathematical models are useful in quantifying parameters related to the pathophysiology of T2D. Here, we extend the minimum model developed to describe the glucose-insulin-glucagon dynamics in the isoglycemic intravenous glucose infusion (IIGI) experiment to the oral glucose tolerance test (OGTT). The extended model describes glucose and hormone dynamics in OGTT including the contribution of the incretin hormones, glucose-dependent insulinotropic polypeptide (GIP), and glucagon-like peptide-1 (GLP-1), to insulin secretion. A new function describing glucose arrival from the gut is introduced. The model is fitted to OGTT data from eight individuals with T2D and eight weight-matched controls (CS) without diabetes to obtain parameters related to insulin sensitivity, β - and α -cell function. The parameters, i.e., measures of insulin sensitivity, α_1 , suppression of glucagon secretion, k_1 , magnitude of glucagon secretion, γ_2 , and incretin-dependent insulin secretion, γ_3 , were found to be different between CS and T2D with P values < 0.002 , < 0.017 , < 0.009 , < 0.004 , respectively. A new rubric for estimating the incretin effect directly from modeling the OGTT is presented. The average incretin effect correlated well with the experimentally determined incretin effect with a Spearman rank test correlation coefficient of 0.67 ($P < 0.012$). The average incretin effect was found to be different between CS and T2D ($P < 0.032$). The developed model is shown to be effective in quantifying the factors relevant to T2D pathophysiology.

NEW & NOTEWORTHY A new extended model of oral glucose tolerance test (OGTT) has been developed that includes glucagon dynamics and incretin contribution to insulin secretion. The model allows the estimation of parameters related to α - and β -cell dysfunction, insulin sensitivity, and incretin action. A new function describing the influx of glucose from the gut has been introduced. A new rubric for estimating the incretin effect directly from the OGTT experiment has been developed. The effect of glucose dose was also investigated.

glucagon secretion; glucagon suppression; incretin effect; insulin resistance; insulin secretion

INTRODUCTION

Glucose homeostasis is maintained through feedback control mechanisms (1, 2). The primary controllers of plasma glucose levels are the hormones insulin and glucagon, secreted by the β - and α -cells, respectively, located in the islets of Langerhans in the pancreas. Insulin secretion occurs in response to glucose-mediated signaling and under postprandial conditions is additionally potentiated by incretin hormone signaling (3, 4). The incretin hormones, glucose-

dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1), are hormones secreted in response to nutrient ingestion by the enteroendocrine K- and L-cells of the gut, respectively (5, 6). GLP-1 potentiates glucose-induced insulin secretion while suppressing glucagon secretion at normal-to-high plasma glucose levels. GIP also potentiates glucose-stimulated insulin secretion, but in contrast to GLP-1, this incretin hormone stimulates glucagon secretion at normal-to-low plasma glucose levels (6). The contribution of incretin signaling to insulin secretion is referred to as the



incretin effect and is critical in postprandial control of plasma glucose levels (4, 6, 7). The regulation of glucagon secretion is complex and driven by both glucose and amino acid sensing by the α -cells in the pancreatic islets and further modulated by insulin, incretins, and somatostatin, a hormone secreted by the delta cells in the islets (8–11).

Type 2 diabetes (T2D) is characterized by elevated fasting as well as postprandial plasma glucose levels as a result of insulin resistance, as well as β - and α -cell dysfunction (12–19). Insulin resistance refers to defective insulin-mediated uptake of glucose by peripheral tissues and/or insulin resistance in the liver where insulin-induced suppression of hepatic glucose production is impaired. Beta-cell dysfunction results in inadequate secretion of insulin and is partly due to reduced incretin effect (6, 7, 17, 20–22). The α -cell dysfunction results in elevated baseline glucagon levels that are inappropriate for the prevailing glycemia compared with what is seen in healthy subjects (23–25). Second, suppression of glucagon secretion in response to glucose stimulus is impaired and less than that observed in normal individuals.

Mathematical modeling may reveal the extent to which each of the aforementioned impairments and dysfunctions contribute to the development of T2D (26–28). A good, fully coupled, physiology-based model that not only includes glucose and insulin, but other regulatory hormones as well, should help estimate the parameters needed to assess the role of α - and β -cell dysfunction as well as insulin sensitivity and incretin effect from one experiment. Parsimonious models (29) are particularly useful in this regard as the parameters can be estimated by fitting the model to readily available data from intravenous or oral glucose tolerance tests (OGTT or IVGTT) without the risk of overfitting. To quantify the extent of insulin resistance and impairment in insulin secretion in T2D, theoretical models describing glucose-insulin dynamics in IVGTT and OGTT have been developed previously (30–34). Many prior models including that of Bergman et al. (31) and Dalla Man et al. (35) only consider glucose and insulin dynamics. Some later models also include glucagon dynamics (36–41). De Gaetano and Hardy (36) include glucagon dynamics but do not explore the role of α -cell dysfunction or develop measures of α -cell dysfunction from the modeling. In other models, the role of insulin in suppressing glucagon secretion has been considered but is limited to that aspect of the dynamics (38). In the minimal model developed by Kelly et al. (37), it is assumed that glucagon secretion only occurs when glucose levels fall below the baseline and that suppression of glucagon secretion is exclusively modulated by insulin. The incretin effect has been included in some models, including those of Dalla Man et al. (42, 43), Morettini et al. (38), Tura et al. (44), Mari et al. (45), and De Gaetano et al. (46, 47) among others (48). In the Dalla Man et al. model of c-peptide secretion during intravenous as well as mixed meal tolerance tests, the incretin potentiated c-peptide secretion rate is assumed to be proportional to glucose-mediated secretion rate. Most of the other models rely on data from both the OGTT and isoglycemic intravenous glucose infusion (IIGI) or OGTT and graded intravenous glucose infusion tests to estimate the incretin effect. In the De Gaetano et al. (47) model, incretins are not explicitly included in the model.

Recently, we developed a delay differential equation-based minimal model of IIGI (49), which included glucagon dynamics and yielded significant information regarding α - and β -cell dysfunction as well as insulin resistance in T2D. The model is extended in this paper to describe the coupled glucose-insulin-glucagon dynamics in an OGTT experiment from which incretin-dependent insulin secretion could also be estimated. Incretin hormones play a significant role in regulating postprandial glycemia and a reliable method of estimating the incretin effect, particularly in individuals with diabetes, where the effect is considerably attenuated and difficult to assess, is important. The model for incretin-dependent insulin secretion is based on the model of Toghiani et al. (46). A new rubric for estimating the incretin effect directly from the OGTT data using the model developed is presented. A new parameterized function for describing glucose appearance from the gut is described. The coupled model allows the estimation of all parameters related to glucose homeostasis such as insulin resistance, glucose- and incretin-dependent insulin secretion, and glucose-dependent glucagon secretion and suppression.

MODEL DEVELOPMENT

Glucose-Insulin-Glucagon Dynamics in OGTT

The coupled ordinary differential equations (ODEs) used here to describe the glucose-insulin-glucagon dynamics in an OGTT represent an extension of the model developed by us to describe IIGI and published in the paper by Subramanian et al. (49). The main differences are the parameterized function describing glucose arrival from the gut and the inclusion of incretin-potentiated insulin secretion in the model. The equations describing the coupled dynamics in an OGTT are given as follows:

$$\frac{dG}{dt} = -(S_G + a_1 I[t])G[t] + a_2 A[t] + R_a[t, p]/V \quad (1)$$

$$\frac{dI}{dt} = -n_1 I[t] + \gamma_1 \Psi(G[t]) + \gamma_{3GLP} GLP-1[t - \tau_{GLP}]G[t] + \gamma_{3GIP} GIP[t - \tau_{GIP}]G[t] \quad (2)$$

$$\frac{dA[t]}{dt} = -n_2 A[t] + \gamma_2 \varphi(G[t - \tau_1]) \quad (3)$$

where G , I , and A represent plasma glucose, insulin, and glucagon concentrations, respectively.

The first equation describes glucose dynamics. The rate constant for glucose-dependent glucose disposal from the plasma, S_G , is kept fixed at population values obtained from the literature (35). The parameters a_1 and a_2 give a measure of insulin sensitivity and glucagon action, respectively. Insulin sensitivity, a_1 , in this model is the rate constant for insulin-dependent glucose disposal. Glucagon action, a_2 , is the rate constant for glucagon-dependent glucose release from the liver. $R_a[t, p]$ is the rate of glucose arrival from the gut post glucose ingestion in the OGTT experiment. Salinari et al. (34), De Gaetano and coworkers (50), and Alskär et al. (51) used a physiology-based model of glucose transit through the stomach and gut to describe $R_a[t, p]$ but several parameters were assigned a priori and some others were

estimated by fitting the model to OGTT data and for that reason not used in this model developed. In the minimal model of OGTT of Cobelli et al. (32, 35, 52, 53), the rate of arrival of glucose from the gut was modeled using a piecewise linear (PWL) model. The problem with the PWL model is that it is not smooth and additionally can take on negative values that are unphysical. The plasma glucose data post-OGTT as well as tracer studies combined with modeling of OGTT experiments (53, 54) indicate that the $R_a[t,p]$ profile has a rapid rising arm followed by a slower trailing arm, similar to a modified Gaussian. A parameterized function would thus be useful in modeling this glucose influx. The skew normal distribution (SND) (55) is a particularly flexible function that shows such a profile. Both the rising arm and the tail of the function can be skewed, in other words, the function can either be left or right skewed making it extremely versatile. The function is smooth, and continuous, and only takes on positive values. In addition, the cumulative distribution function (CDF) of the distribution is readily calculated. A skew normal distribution was used to model $R_a[t,p]$ with parameters $p \rightarrow \{\mu, \sigma, \alpha\}$:

$$\text{SND}(\mu, \sigma, \alpha, t) = \frac{2}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} \int_{-\infty}^{\alpha\frac{(t-\mu)}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \quad (4)$$

Depending on whether the glucose profile post-OGTT had a single peak or two peaks, one or two SNDs were used to fit the data.

$$R_a[t,p] = b_1 \text{SND}(\mu_1, \sigma_1, \alpha_1, t) + b_2 \text{SND}(\mu_2, \sigma_2, \alpha_2, t) \quad (5)$$

The integral of the glucose arrival rate is constrained to be equal to a fraction of the ingested glucose as described in the paper by Dalla Man et al. (35) as follows:

$$\int_0^T R_a[t,p] dt = \frac{D \times f}{\text{BW}} \quad (6)$$

D is the ingested glucose dose in mg, f is the fraction of the dose that is assumed to be absorbed and available after transiting the liver (56) and BW is the body weight in kg. V in Eq. 1 is the volume of distribution per unit body weight (dL/kg).

Equation 2 describes insulin dynamics. The insulin clearance rate is given by n_1 . The rate constant for glucose-dependent insulin secretion is given by γ_1 . The incretin-augmented insulin secretion is described by the last term and γ_3 is the rate constant for GIP or GLP-1-potentiated insulin secretion.

The glucose-dependent insulin dose-response curve ψ was modeled by the Hill equation. The parameters of the Hill equation were determined by fitting islet level data from normal individuals obtained from the paper by Rorsman and co-workers (57). The constant K was determined to be 17.0 mM and the Hill coefficient, h , 1.27. There are two in vivo methods for determining insulin secretion in response to glucose. The first uses deconvolution of c-peptide dynamics to estimate insulin secretion rates in graded glucose infusion experiments (58–60). A dose-response curve is then estimated from average values of the insulin secretory rate in response to the average plasma glucose level during each

glucose infusion step of 30–40 min. This dose-response curve is thus only an approximation of the actual dose-response curve that ideally should be estimated at fixed values of plasma glucose. Also, as the plasma glucose levels never reach very high levels during the graded infusion in normal subjects, the dose-response curve was only estimated up to 180 mg/dL and found to be linear in the plasma glucose range achieved. The second method used to estimate the dose-response curve is the hyperglycemic clamp experiment repeated at multiple glucose clamp values (61). The first-phase [area under the curve (AUC) insulin first 10 min of the experiment] and second-phase insulin secretion (60-min insulin value) are then plotted against clamped glucose values to obtain the dose-response curves. Alternatively, the c-peptide values were used. The dose-response curves were then fit to the Hill equation and the parameters, K and the Hill coefficient determined. The value for K (glucose—30 min) estimated using insulin data was 16.9 mM, very similar to the value of 17 mM determined from islet level studies. A Hill coefficient of 1.5 was found to be adequate to describe the dose-response curve and compared well with the value of 1.27 obtained from islet level studies. The K value determined from c-peptide data was lower at 13.3 mM. In the paper, plasma insulin levels or c-peptide values were directly used to determine the dose-response relationship with glucose. Using c-peptide deconvolution to estimate insulin secretion rates as in the first in vivo method would be more appropriate. In conclusion, the results from islet level studies compare well with the second in vivo method of estimating the dose-response curves. The advantages of the islet level data are that there are no confounding effects of insulin degradation or clearance by the liver and were thus used in our paper

$$\psi = \frac{1.5(G[t])^h}{K^h + (G[t])^h} \quad (7)$$

As the incretin effect is significant after oral ingestion of carbohydrates, the contribution of GIP and GLP-1 was also included in the modeling. Insulin secretion in response to GIP and GLP-1 is modeled in a glucose-dependent manner (62) and is taken to be first order in incretin and glucose (46, 63). To reduce the number of equations and parameters, the contribution of incretins to the dynamics was included by introducing it as an interpolated function in the insulin dynamics equation (Eq. 2). The effect of the incretins on glucagon secretion is not included in this model as it has been shown that the stimulatory effect of GIP and the inhibitory effect of GLP-1 may counterweigh each other (64, 65).

Equation 3 describes glucagon dynamics. The glucagon clearance rate is given by n_2 . The glucagon dose response is modeled as decreasing exponentially in response to increasing glucose levels (57, 66). From an examination of the data from OGTT tests and preliminary modeling, it was apparent that the recovery of glucagon secretion, as glucose levels return to normal, is very slow. This was also observed when modeling IIGI (49). Thus, the nature of the glucagon response is assumed to change during the recovery phase as glucose levels start dropping after reaching a maximum. Although the mechanism underlying this behavior, which is hysteresis-like, is uncertain, the use of two different exponential responses to glucose levels captured the dynamics

reasonably well as seen in RESULTS. The glucagon dose response, ϕ , was thus modeled as given as follows:

$$\phi(G[t]) = e^{-k_1 G[t]} (H[t] - H[t - t_{\text{hyst}}]) + (e^{-k_2 G[t]} + \text{yshift}) (H[t - t_{\text{hyst}}]),$$

$$\text{yshift} = e^{-k_1 G_{\text{hyst}}} - e^{-k_2 G_{\text{hyst}}} \quad (8)$$

where H is the Heaviside function, t_{hyst} is the time at which the value of plasma glucose concentration reaches a maximum, G_{hyst} , post glucose ingestion in the OGTT experiment. A shift is introduced so the two curves intersect at the point $\{G_{\text{hyst}}, t_{\text{hyst}}\}$. The value of k_1 obtained from fitting the average islet level data from normal individuals with a single exponential was 0.25 mM^{-1} . Figure 1 shows the glucagon response as a function of glucose concentration. As glucose levels rise, glucagon secretion falls exponentially. The recovery follows a different path. The effect on glucagon dynamics is a delayed return to baseline values.

Estimation of the Incretin Effect from OGTT

In this paper, we present a method of estimating the contribution of the incretins GLP-1 and GIP to insulin secretion directly from the OGTT data using the model developed as follows:

Step 1: The model in Eqs. 1–3, including either the GLP-1 or GIP-induced insulin secretion, is first fit to the data and the rate of influx of glucose $R_a[t, p]$ and other parameters determined.

Step 2: Equation 2 is modified to include only the glucose-dependent insulin secretion term. The model is then simulated with the parameters estimated in step 1 but without including the GLP-1 or GIP contribution to insulin secretion, i.e., dropping the third and fourth term in Eq. 2. The influx of glucose from the gut $R_a[t, p]$ is the same as that calculated in step 1. The area under the insulin curve (AUC) which gives the glucose-dependent insulin response (AUC GDIR) is calculated.

Step 3: The model is simulated without the contribution of the glucose-dependent insulin response (GDIR), i.e., dropping the second term in Eq. 2. The influx of glucose from the gut is the same as that calculated in step 1. Either the GIP or

GLP-1 contribution to the insulin response is included in the simulation. The area under the insulin curve that gives the GLP-1 (AUC GLPIR) or GIP (AUC GIPIR) potentiated insulin response is calculated.

The incretin effect is then given by:

$$\frac{\text{AUC GLPIR or AUC GIPIR}}{\text{AUC GDIR} + (\text{AUC GLPIR or AUC GIPIR})} \times 100\% \quad (9)$$

In the new method presented here to calculate the incretin effect, the same influx of glucose from the gut occurs in all three steps. The incretin or glucose mediated β -cell response is switched off in steps 2 and 3, respectively, to estimate their individual contributions to insulin secretion.

PARAMETER ESTIMATION AND STATISTICS

The coupled Eqs. 1–3 were fit using one of two methods: 1) In the first method, the parameters of the coupled equations (Eqs. 1–3) were fit simultaneously to the data using the nonlinear least squares and Brent's Principal Axis algorithm (NonLinearModelFit[], Wolfram Mathematica 12.0) (67). The parameters determined from the fit were a_1 , a_2 , γ_1 , γ_2 , γ_3 , μ , σ , α . The parameters related to glucagon suppression, k_1 and k_2 were adjusted manually around the average value determined for normal human individuals from islet level studies. The time delays in glucagon secretion and/or incretin secretion, τ_1 , $\tau_{\text{GLP-1}}$, and τ_{GIP} , were also adjusted manually.

2) In the second method, the glucose equation (Eq. 1) was first fitted to the OGTT glucose time series. The glucagon and insulin data were introduced as interpolated functions, $I[t]$ and $A[t]$. This first step yielded the parameters a_1 , a_2 , μ , σ , α . In the second step, the three coupled equations (Eqs. 1–3) were solved simultaneously with the parameters determined in the first step kept fixed. In this second step, all the remaining parameters k_1 , γ_1 , γ_2 , τ_1 , $\tau_{\text{GLP-1}}$, and τ_{GIP} were determined. The glucagon recovery parameter, k_2 , was adjusted manually. When modeling control subjects, a sum of two SNDs were required to determine $R_a[t, p]$, this required the parameters b_1 , b_2 , μ_1 , μ_2 , α_1 , α_2 to be estimated from the fit, whereas σ_1 and σ_2 were tuned manually. In individuals with diabetes, one SND was sufficient to describe $R_a[t, p]$ and three parameters μ , σ , and α , were estimated in that case. The inbuilt SND function "SkewNormalDistribution" in Mathematica was used in the modeling. The nonlinear least squares fit was done using the Levenberg–Marquardt algorithm (NonlinearModelFit[], Wolfram Mathematica 12.0) (67).

The parameters n_1 , n_2 , S_g , V , and f were kept fixed at values of 0.14 min^{-1} , 0.08 min^{-1} , 0.014 min^{-1} , 1.35 dL/kg , and 0.86 , respectively, and were obtained from the literature (35, 49). The parameters of the Hill equation in Eq. 7 were kept fixed at $K=17.0 \text{ mM}$ and the Hill coefficient at $h=1.27$, values determined from islet level data (57).

As both incretins contribute to insulin secretion in a similar glucose-dependent manner in Eq. 2, the joint or combined contribution of GIP and GLP-1 could not be uniquely identified from fitting the model. Thus, Eqs. 1–3 with the contribution of either GLP-1 or GIP was fit to the data. The incretin effect was then estimated for GLP-1 and GIP separately. The average of the two values was then taken to be an estimate of the combined incretin effect.

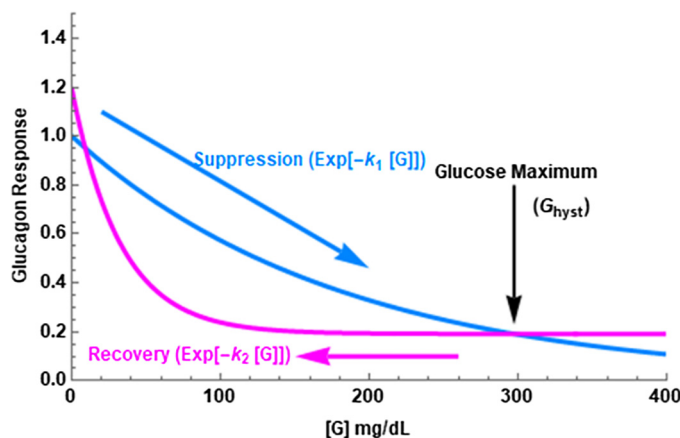


Figure 1. The hysteresis-like behavior of the glucagon response is depicted in the figure. As glucose levels rise post glucose challenge, the glucagon response follows the blue curve till glucose levels reach the maximum (G_{hyst}). As glucose levels return to baseline, the glucagon response follows the magenta curve.

The OGTT data from eight individuals with diabetes and eight weight-matched controls without diabetes was used to parameterize the model. Comparison of the model without the incretin effect and when including the incretin effect was made using the Akaike information criterion (AIC) (68–70), as well as AIC corrected for small sample size (AICc). Though the absolute value of AIC or AICc is not meaningful, a lower relative value of AIC or AICc, when comparing models, indicates a better fit. If the difference in AICc (ΔAICc) between two models is 2 or greater, the model with the lower AICc can be considered the better model based on the calculation of Akaike weights (69). In controls (CS), the AIC and AICc were calculated for the full model. In T2D, the AIC and AICc were calculated for the insulin fit.

To assess the extent of compensation in β -cell function in response to decline in insulin sensitivity, the disposition index (DI) (71–74) was calculated from the area under the insulin and glucose curve post glucose challenge and the insulin sensitivity parameter, a_1 , estimated by fitting the data, as follows:

$$\text{DI} = \frac{\text{AUC}_{I_{0-240}}}{\text{AUC}_{G_{0-240}}} \times a_1 \quad (10)$$

The Wilcoxon signed-rank test (75) at the 5% significance level was used to make the comparisons between estimated parameters within a group and the Mann–Whitney U test (76) at the 5% significance level was used for between-group comparisons.

EXPERIMENTAL METHODS

The details of the experimental protocols followed, and data collection methods are presented in detail in the papers by Bagger et al. (17, 25). The protocol was approved by the Scientific-Ethical Committee of the Capital Region of Denmark (Registration No. H-A-2007- 0048), the Danish Data Protection Agency (Registration j.nr. 2007-41-1058) and at www.ClinicalTrials.gov (IDNCT00529048). The study was conducted according to the principles of the Helsinki Declaration II.

Subjects

Eight individuals (3 males) with T2D [mean age, 57 (range: 40–75) yr; body mass index (BMI) 29 (25–34) kg/m²; HbA1c 7.0 $\pm$ 0.3% [53 $\pm$ 2 mmol/mol]; duration of diabetes 8 (6–36) mo] and 8 gender-, age-, and BMI-matched healthy control [age, 57 (38–74) yr; BMI, 29 (26–33) kg/m²; HbA1c 5.4 $\pm$ 0.1% [36 $\pm$ 1 mmol/mol]] were studied. All individuals with T2D were diagnosed based on the criteria of the World Health Organization (77).

Participants were subjected to OGTT after an overnight 10-h fast. The participants ingested 75 g of glucose dissolved in 300 of g water and blood samples were drawn 15, 10, 0 before and 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 90, 120, 150, 180, 240 min after ingestion of glucose. The analytical methods used to estimate glucose, insulin, glucagon, and incretin concentrations are described in the study by Bagger et al. (17, 25). The experiments were repeated on the same individuals with amount of glucose of 25 g and 125 g.

RESULTS

As described in PARAMETER ESTIMATION AND STATISTICS, two methods were used to fit the model to the data. *Method 1*, where the coupled equations (Eqs. 1–3) are simultaneously fit to the data in one step, worked well for the individuals with diabetes as the glucose arrival from the gut could be described by one parameterized skew normal distribution function. In CS, the glucose arrival shows a biphasic profile and needs to be modeled with the sum of two skew normal distributions. This increases the number of parameters needed to be estimated and *method 2* was found to be more efficient both in terms of time required for the runs and the precision with which the parameters could be estimated. A comparison of the results obtained using the two methods in individuals with diabetes is presented in the Supplemental Table S1. As the differences between the two methods were not statistically significant, in the rest of the paper, only the results obtained using *method 2* are presented.

This section is organized as follows: First results from fitting the model equations (Eqs. 1–3) to data from the 75 g OGTT on CS and individuals with T2D are presented. This is followed by results from the 25 g and 125 g OGTT data. A comparison of the results from the three experiments is then presented.

75 g OGTT

Glucose arrival ($R_a(t)$).

Model equation (Eq. 1) was fitted to the data from eight CS and eight individuals with T2D according to *step 1, method 2*. In this step, the parameters, a_1 , a_2 , γ_1 , γ_2 , γ_3 , μ , σ , α were determined. In CS, two parametrized skew normal distributions were used to describe $R_a(t)$ as in Eq. 5. The coefficient of determination values for the fits were uniformly high (adjusted $R^2 > 0.99$).

The glucose arrival profiles for CS and T2D obtained using the SND model are presented in Fig. 2. The glucose arrival from the gut is bimodal in CS, whereas it is primarily unimodal in T2D. Second, the glucose arrival is much more rapid in CS relative to individuals with T2D.

As a new parametrized function, the skew normal distribution, is used to characterize glucose arrival, a brief comparison will also be made with the PWL model of Cobelli and coworkers (53). In Supplemental Fig. S1, the comparison with the $R_a(t)$ profiles obtained using the PWL and SND models is presented. Though the profiles are similar, the SND-based $R_a(t)$ profiles are smooth and do not take on unphysiological negative values.

Indirect validation of the glucose arrival profiles was obtained by comparing the $R_a(t)$ curves with the glucose infusion profiles in the companion IIGI experiment (17). A direct validation would be from tracer studies (53, 78), which were not carried out in this cohort. In Supplemental Fig. S2, A and B, the glucose arrival calculated by fitting OGTT data using *method 2*, and the SND model, is compared with the glucose arrival from the corresponding (IIGI) experiment in both CS and individuals with T2D. The glucose arrival profile in the IIGI experiment mirrors the estimated $R_a(t)$ from the OGTT reasonably well though the amount of glucose infused in IIGI is different from the amount ingested during the OGTT.

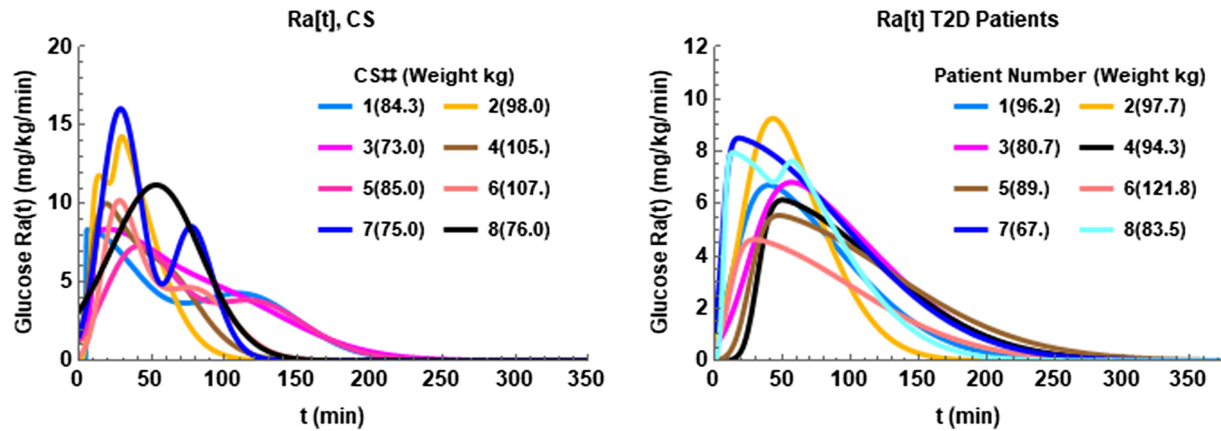


Figure 2. The glucose arrival profiles, $R_a[t]$, as determined from fitting the model to seven controls (CS) and eight individuals with type 2 diabetes (T2D).

CS. The fits obtained from *step 2, method 2*, are shown in Figs. 3 and 4 for CS 2 and CS 4, respectively. In Fig. 3A and Fig. 4A, the fit obtained without including incretin effects is presented. Though the initial rise in insulin secretion is modeled well, the entire secretory profile is not captured well. In Fig. 3, B and C and Fig. 4, B and C, the fits obtained by including GLP-1- or GIP-induced insulin secretion in

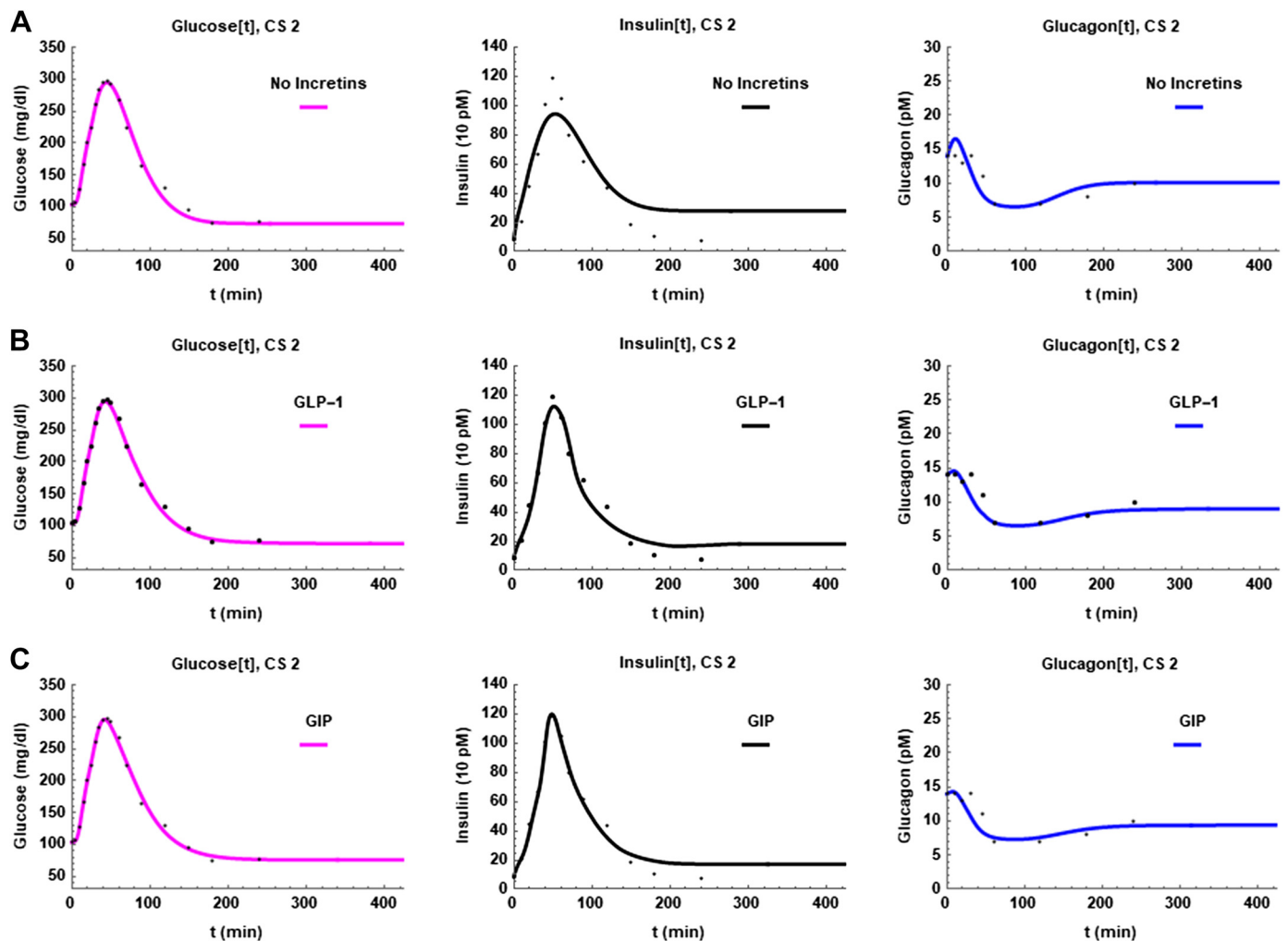


Figure 3. Representative fit of the model to data from controls (CS) 2 is presented. A: fit of the oral glucose tolerance test (OGTT) model without including incretin effects to glucose (magenta curve), insulin (black curve), and glucagon (blue curve) data (black dots). B and C: fits including glucagon-like peptide-1 (GLP-1) or glucose-dependent insulintropic polypeptide (GIP) effect, respectively, in addition to glucose-stimulated insulin secretion.

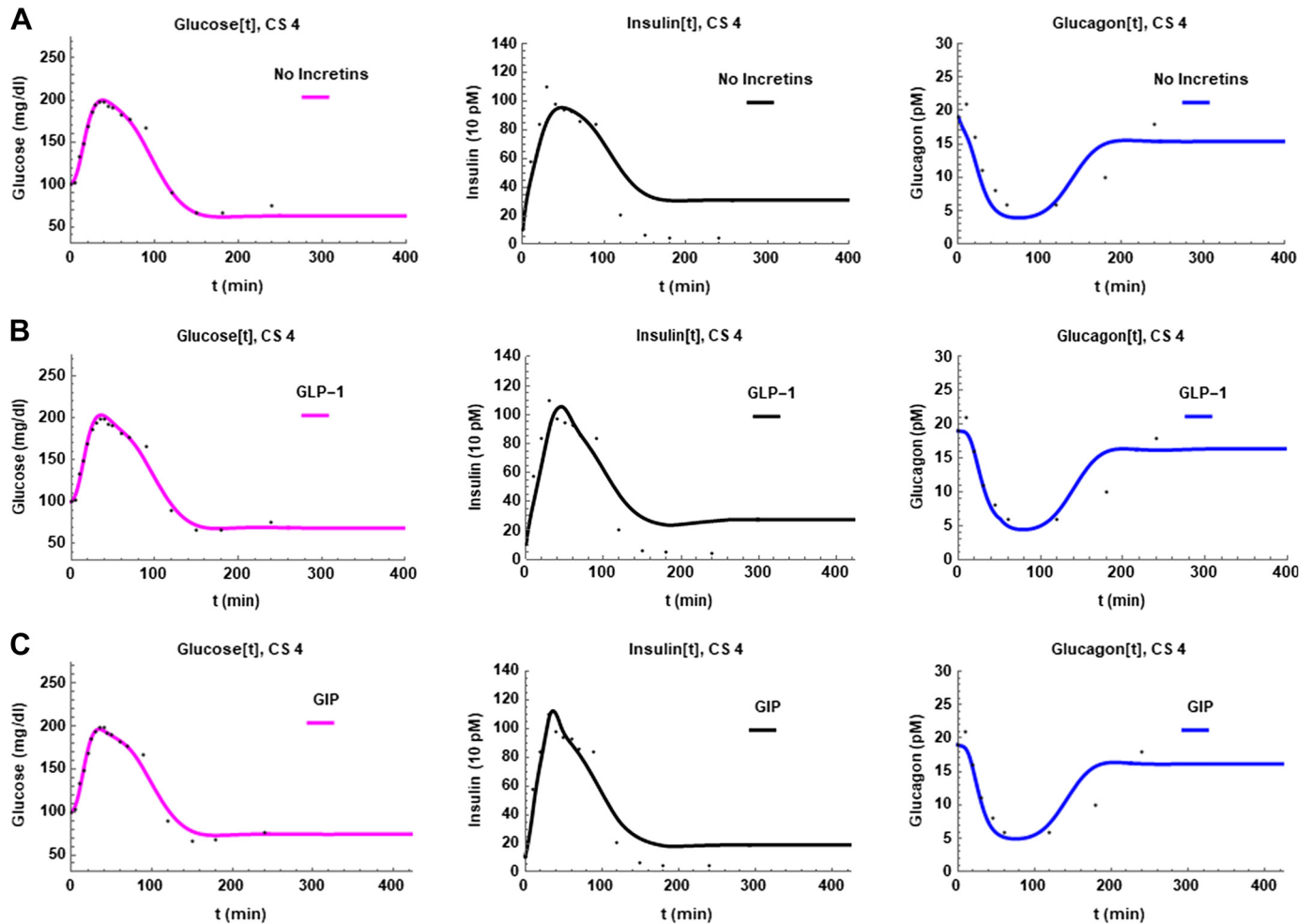


Figure 4. Representative fit of the model to data from controls (CS) 4 is presented. *A*: fit of the oral glucose tolerance test (OGTT) model without including incretin effects to glucose (magenta curve), insulin (black curve), and glucagon (blue curve) data (black dots). *B* and *C*: fits including glucagon-like peptide-1 (GLP-1) or glucose-dependent insulinotropic polypeptide (GIP) effect, respectively, in addition to glucose-stimulated insulin secretion.

addition to the glucose-stimulated insulin secretion is presented. Inclusion of the incretin contribution in the model gives a much better description of insulin secretion in OGTT. Model comparison was made using AIC and AICc values both of which were much lower (average $\Delta\text{AICc} = 24$ and 14 for GIP and GLP-1 fits, respectively) when the incretin contribution was included, which indicates it is the better model of the two.

The mean parameters, SE, and range of parameters obtained by fitting the model including either the GLP-1- or GIP-dependent secretion terms in Eqs. 1–3 are shown in Table 1. A good fit was not obtained for CS 3 and this subject was excluded from further consideration. The coefficient of determination, R^2 , value for all the fits for *step 1* and *step 2* were >0.98 . The standard errors were uniformly low and the P values <0.05 for all parameters were estimated.

Table 1. Parameters obtained by fitting the model equations (Eqs. 1–3) to data from seven CS using method 2

Parameter	CS			
	Mean (SE)		Range	
	GLP-1	GIP	GLP-1	GIP
$a_1 (\times 10^{-4}) (10 \text{ pM} \cdot \text{min})^{-1}$	5.0 (0.92)	5.0 (0.92)	1.6–7.6	1.6–7.6
$a_2, \text{mg/dL} (10 \text{ pM} \cdot \text{min})^{-1}$	0.2 (0.029)	0.2 (0.029)	0.08–0.31	0.08–0.31
k_1, min^{-1}	0.25 (0.041)	0.30 (0.076)	0.073–0.45	0.07–0.72
k_2, min^{-1}	0.49 (0.030)	0.49 (0.030)	0.35–0.55	0.35–0.55
$\gamma_1, 10 \text{ pM} \cdot \text{min}^{-1}$	5.6 (1.3)	5.0 (1.5)	1.7–12.	1.0–11.
$\gamma_2, \text{pM} \cdot \text{min}^{-1}$	2.6 (0.56)	2.8 (0.64)	1.4–5.8	1.5–6.2
$\gamma_3, (\text{mg/dL} \cdot \text{min})^{-1}$	0.0068 (0.0012)	0.0031 (0.00059)	0.0035–0.012	0.0013–0.0055

The mean values, standard error of the mean (SE), and range of the data are presented. CS, controls; GIP, glucose-dependent insulinotropic polypeptide; GLP-1, glucagon-like peptide-1.

The insulin sensitivity parameter a_1 and glucagon action parameter a_2 are determined in *step 1 of method 2*. The a_1 values range from 1.6×10^{-4} to 7.6×10^{-4} . The a_2 values range from 0.08 to 0.31. The average value of the glucagon suppression parameter k_1 obtained from the GIP fit (0.30) and GLP-1 (0.25) is close to the population average value of 0.25 obtained by fitting data from pancreatic islets isolated from human individuals without diabetes (49, 57). The parameters, k_1 , γ_1 and γ_2 , determined using either GIP or GLP-1 are found to be similar. The range for these parameters is wide but the SE is relatively low indicative of outliers. As glucagon suppression showed hysteresis-like behavior during the course of the dynamics, the recovery of glucagon levels to baseline had to be described by a second exponential term and the parameter associated with it is k_2 . The mean recovery constant was found to be 0.49 and showed less variation than the suppression constant k_1 . The constant γ_3 gives a measure of the incretin-potentiated insulin secretion. The values range from 0.0035 to 0.012 with a mean of 0.0068 for GLP-1 and 0.0013–0.0055 with a mean of 0.0031 for GIP. Time delays in incretin-potentiated insulin release (τ_{GLP} and

τ_{GIP}) and the glucagon response term (τ_1) were observed in only two CS and averaged 15 min and 5 min, respectively.

T2D.

The fits obtained using *method 2* as described in the METHODS section are seen in Fig. 5 for T2D subject 1 and in Fig. 6 for T2D subject 8. The fits in Fig. 5A and Fig. 6A (No Incretins) were from solutions to Eq. 1–3 without the inclusion of the GIP- and GLP-1-dependent insulin secretion terms in Eq. 2. In Fig. 5B and Fig. 6B (GLP), the GLP-1-dependent insulin response term is included but not the contribution from GIP. In Fig. 5C and Fig. 6C, the GIP-dependent insulin response term is included but not GLP-1. The AIC and AICc values obtained for the insulin fit were lower when the incretin effect was included in all subjects except *patient 8* where it was comparable (average $\Delta\text{AICc} = 5$ and 7 for GIP and GLP-1 fits, respectively). *Patient 3* showed no incretin effect.

The mean parameters, SE, and range of parameters obtained by fitting the model including either the GLP-1- or GIP-dependent secretion terms in Eqs. 1–3 are shown in Table 2.

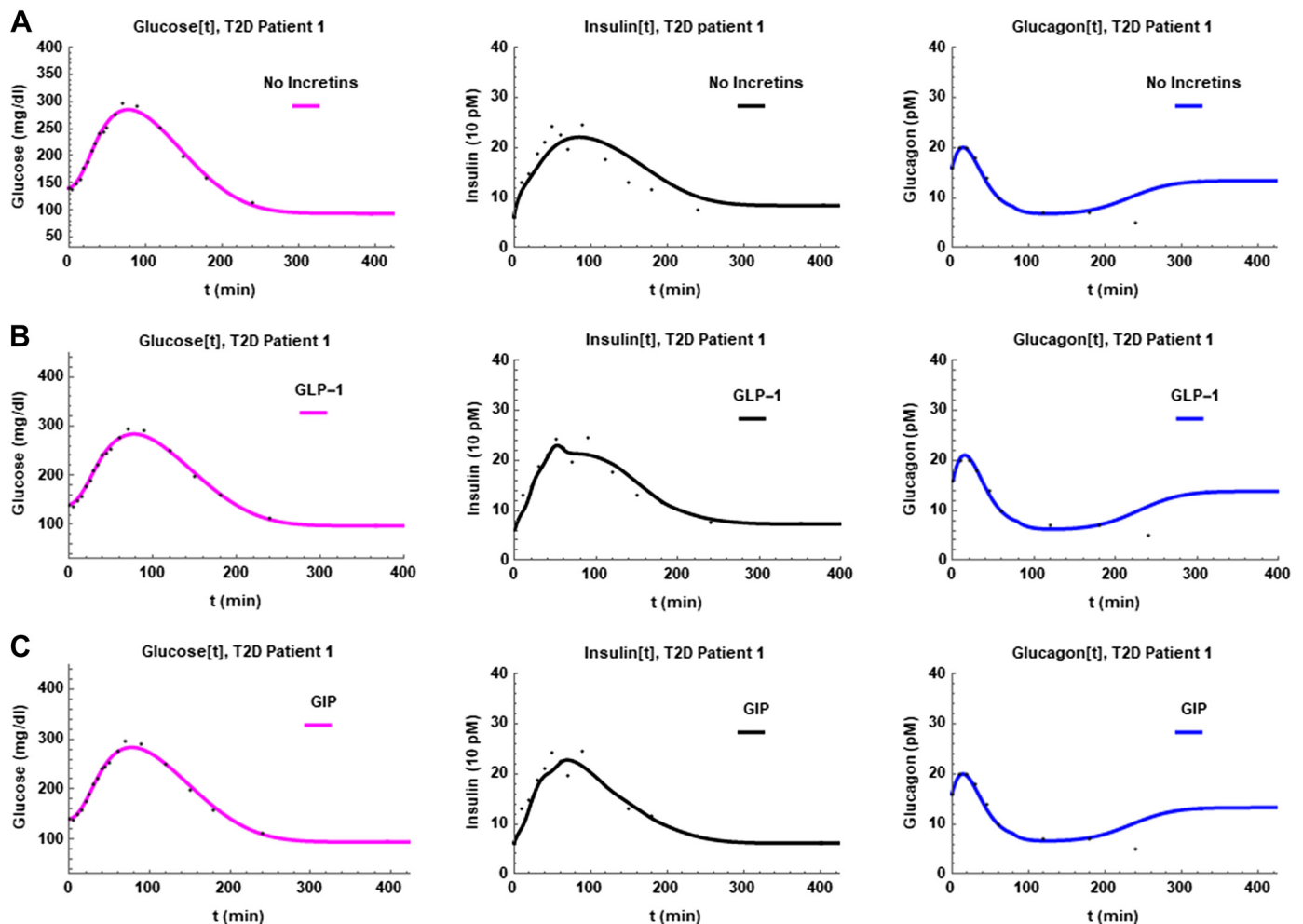


Figure 5. Representative fit of the model to data from type 2 diabetes (T2D) *patient 1* is presented. A: fit of the oral glucose tolerance test (OGTT) model without including incretin effects to glucose (magenta curve), insulin (black curve), and glucagon (blue curve) data (black dots). B and C: fits including glucagon-like peptide-1 (GLP-1) or glucose-dependent insulinotropic polypeptide (GIP) effect, respectively, in addition to glucose-stimulated insulin secretion.

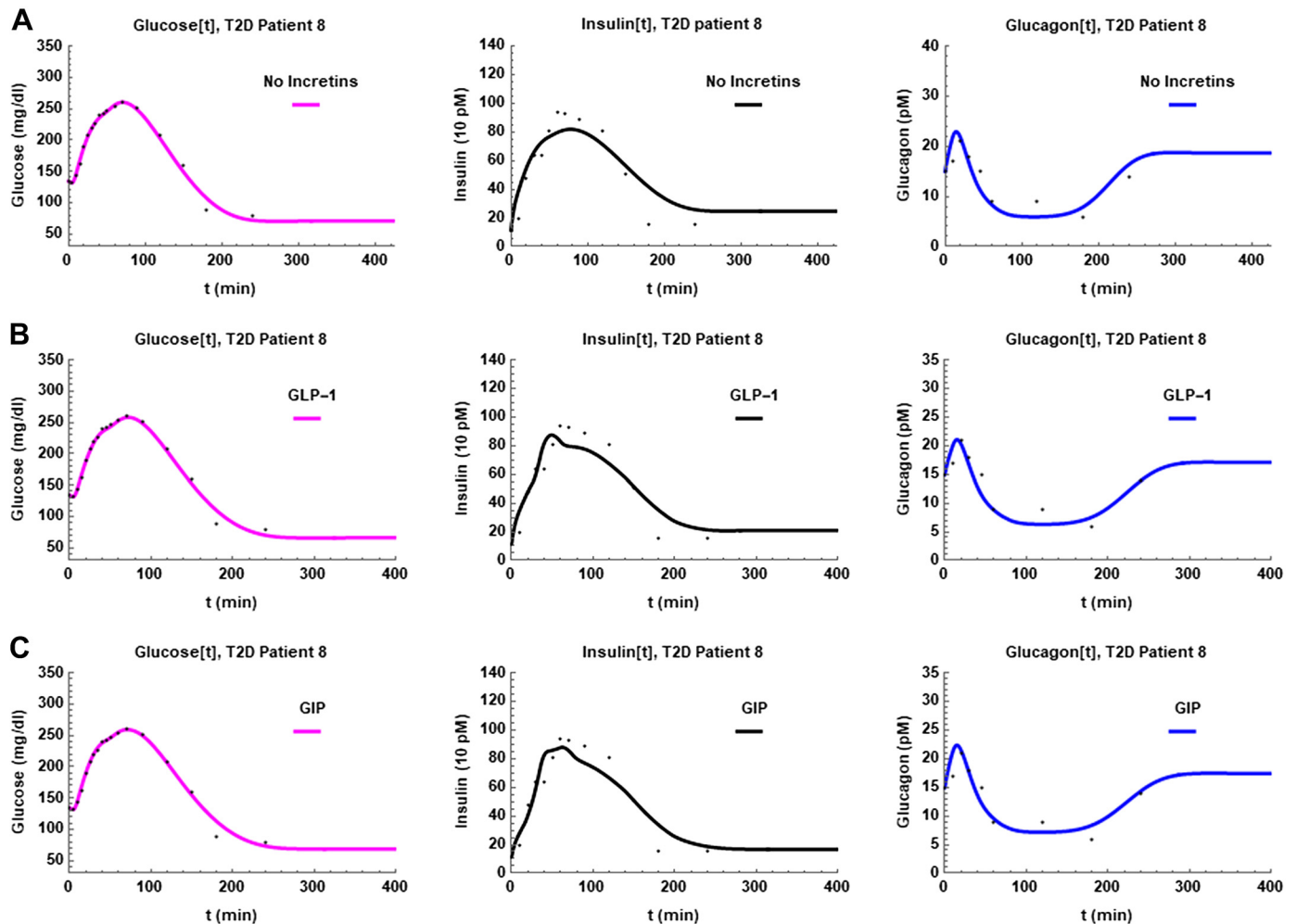


Figure 6. Representative fit of the model to data from type 2 diabetes (T2D) patient 8 is presented. *A*: fit of the oral glucose tolerance test (OGTT) model without including incretin effects to glucose (magenta curve), insulin (black curve), and glucagon (blue curve) data (black dots). *B* and *C*: fits including glucagon-like peptide-1 (GLP-1) or glucose-dependent insulintropic polypeptide (GIP) effect, respectively, in addition to glucose-stimulated insulin secretion.

The coefficient of determination, R^2 , value for all the fits for *step 1* and *step 2* were >0.98 in T2D 1–8. The standard errors were uniformly low and the P values <0.05 for all parameters were estimated.

The insulin sensitivity parameter a_1 and glucagon action parameter a_2 are determined in *step 1* of *method 2*. The a_1 values range from 0.52×10^{-4} to 1.7×10^{-4} with a mean of 1.1×10^{-4} . The a_2 values range from 0.061 to 0.54 with a mean of 0.26. The average value of the glucagon suppression parameter k_1 obtained from the GLP-1 fit (0.15) and GIP (0.15) is much less than the population average value of 0.25 obtained by fitting data from pancreatic islets isolated from human subjects without diabetes. The parameters, k_1 , γ_1 and γ_2 , determined using either GLP-1 or GIP are found to be similar. The range for these parameters is wide but the SE is relatively low indicative of outliers. The mean recovery constant, k_2 , was found to be 0.56. The measure of glucose-stimulated insulin secretion, γ_1 , ranged from 1.6 to 10.0 with a mean of 4.0 for GLP-1 fit and a range of 3.0–11.0 with a mean of 4.3 for the GIP fit. The constant γ_3 gives a measure of the incretin-potentiated insulin secretion. The

values range from 0.–0.0052 with a mean of 0.0017 for GLP-1 and range of 0.–0.0014 with a mean of 0.00066 for GIP. Time delays in incretin-potentiated insulin release, τ_{GLP} and τ_{GIP} , of 15 min were observed in only one patient with T2D. Time delays in glucagon response to glucose challenge, τ_1 , were observed in most individuals with T2D. The average delay was 5 min, and the maximum was 12 min.

Comparison between CS and individuals with T2D.

To assess which aspects of the coupled dynamics contributed to the development of T2D, a comparison between the parameters obtained for the CS and individuals with T2D was made. The Mann–Whitney U test was used to determine which parameters were significantly different between the two groups.

Comparison of the parameters obtained from modeling the 75 g OGTT experiment on eight individuals with T2D and seven CS is presented in Table 3. Significant differences were found in the parameter related to insulin sensitivity, a_1 , with the mean value of 0.0005 and 0.00011 in CS and individuals with T2D, respectively. The mean value of

Table 2. Parameters obtained by fitting the model equations (Eqs. 1–3) to data from the 75 g OGTT on eight T2D individuals using method 2

Parameter	T2D			
	Mean (SE)		Range	
	GLP-1	GIP	GLP-1	GIP
$\alpha_1 (\times 10^{-4}) (10 \text{ pM} \cdot \text{min})^{-1}$	1.1 (0.16)	1.1 (0.16)	0.52–1.7	0.52–1.7
$\alpha_2, \text{mg/dL} (10 \text{ pM} \cdot \text{min})^{-1}$	0.26 (0.063)	0.26 (0.063)	0.061–0.54	0.061–0.54
k_1, mM^{-1}	0.15 (0.01)	0.15 (0.01)	0.12–0.2	0.12–0.20
k_2, mM^{-1}	0.56 (0.027)	0.56 (0.027)	0.45–0.65	0.45–0.65
$\gamma_1, 10 \text{ pM} \cdot \text{min}^{-1}$	4.0 (1.2)	4.3 (1.2)	1.6–10.	3.0–11.
$\gamma_2, \text{pM} \cdot \text{min}^{-1}$	5.0 (0.65)	5.2 (0.74)	3.0–7.8	1.2–13.
$\gamma_3, (\text{mg/dL} \cdot \text{min})^{-1}$	0.0017 (0.00054)	0.00066 (0.00021)	0.0–0.0052	0.0–0.0014
τ_1, min	4.4 (1.3)	4.5 (1.3)	0.0–10.0	0.0–12.

The mean values, standard error of the mean (SE), and range of the data are presented for the glucose-dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) fits.

the glucagon suppression parameter, k_1 , was significantly different between the two groups and much lower in T2D (0.15) relative to CS (0.25). The mean value of the measure of glucagon secretion, γ_2 , was again significantly different between CS (2.6) and T2D (5.0). The incretin-dependent insulin secretion, γ_3 , was significantly different between CS (0.0068) and T2D (0.0017) for the GLP-1 fit and CS (0.0031) and T2D (0.00066) for the GIP fit. Interestingly, the parameter γ_1 , which is a measure of glucose-stimulated insulin secretion was not found to be significantly different between CS and individuals with T2D. In addition, the glucagon action parameter and the glucagon recovery parameter were not found to be significantly different between the two groups.

Estimation of the incretin effect from OGTT.

The insulin secretion profile obtained in step 1, step 2, and step 3 in *Estimation of the Incretin Effect from OGTT* is plotted in Fig. 7, A–D for CS and Fig. 8, A–D for individuals with T2D. The glucose-dependent insulin response (step 2, GDIR, blue curve), the incretin-dependent insulin response (step 3, GLP-1IR or GIP, yellow curve), the original fit from step 1 including both glucose- and incretin-dependent insulin response terms (step 1, Ofit, purple curve), and the sum of

GIP or GLP-1IR and GDIR is represented by the magenta curve. In the CS, the sum of GDIR and GLP-1IR or GIP (magenta curve) is higher than in the original fit. In individuals with T2D, the magenta curve and the purple curve coincide. These plots also reveal that the characteristic shape of the insulin profile can largely be attributed to the incretin-dependent insulin response. This is true even in the case of the individuals with T2D in this cohort where the incretin effect is lower.

The incretin effect calculated using Eq. 8 and following the steps described in *Estimation of the Incretin Effect from OGTT* is presented in Table 4. The calculated incretin effect from the GIP and the GLP-1 fit are presented in the 2nd and 3rd columns. As the actual incretin effect is due to both GLP-1 and GIP, the average of the two estimates from the GLP-1 and GIP fits is presented in the 4th column. The experimentally determined incretin effect estimated in two different ways (17) is presented in the fifth column. In method 1, absolute insulin values were used and in method 2 baseline insulin values were subtracted when determining the incretin effect.

In Table 5, comparisons between the calculated incretin effect in CS and T2D as well as the experimentally determined incretin effect are presented. The calculated incretin effects from the GLP and GIP fits are not statistically significantly different in the CS individuals (Wilcoxon signed-rank test P value < 0.93). The mean values are 56% and 59%, respectively. In individuals with T2D with an incretin effect, the calculated effect from the GIP fit is significantly different from that obtained from the GLP fit in individuals with T2D (Wilcoxon signed-rank test P value < 0.022). The mean values of the GLP and GIP effect are 36% and 24%, respectively, in the individuals with T2D.

The incretin effect is found to be significantly different between CS and individuals with T2D whether comparing the effect from the GIP fit, GLP-1 fit, or the average value (Mann-Whitney U test P value < 0.0014 , 0.024, 0.032, respectively).

The calculated incretin effect from the GIP fit, the GLP-1 fit, the average calculated value, and the experimentally determined incretin effect are not found to be significantly different (Mann-Whitney U test P values: < 0.52 , 0.60, 0.48, respectively). In Fig. 9, A and B, the calculated estimate of the incretin effect from the OGTT from the GIP-, GLP-1- fit, and the average values are plotted against the experimentally determined incretin effect. The linear correlation is

Table 3. Comparison of the parameters obtained from modeling the 75 g OGTT experiment on 8 individuals with T2D and 7 individuals without diabetes

Parameter	CS vs. T2D		
	Mean CS	Mean T2D	P Value
$\alpha_{1\text{GLP/GIP}} (\times 10^{-4})$	5.0	1.1	< 0.0021
$\alpha_{2\text{GLP/GIP}}$	0.2	0.26	0.95
$k_{1\text{GLP}}$	0.25	0.15	0.017
$k_{1\text{GIP}}$	0.3	0.15	0.017
$k_{2\text{GLP}}$	0.49	0.56	0.16
$k_{2\text{GIP}}$	0.49	0.56	0.16
$\gamma_{1\text{GLP}}$	5.6	4.	0.18
$\gamma_{1\text{GIP}}$	5.	4.3	0.86
$\gamma_{2\text{GLP}}$	2.6	5.	0.0091
$\gamma_{2\text{GIP}}$	2.8	5.2	0.024
$\gamma_{3\text{GLP}}$	0.0068	0.0017	0.0038
$\gamma_{3\text{GIP}}$	0.0031	0.00066	0.0021

Significant differences were found in parameters related to insulin resistance (α_1), glucagon suppression (k_1), glucagon secretion (γ_2), and incretin-dependent insulin secretion (γ_3). CS, control; T2D, type 2 diabetes.

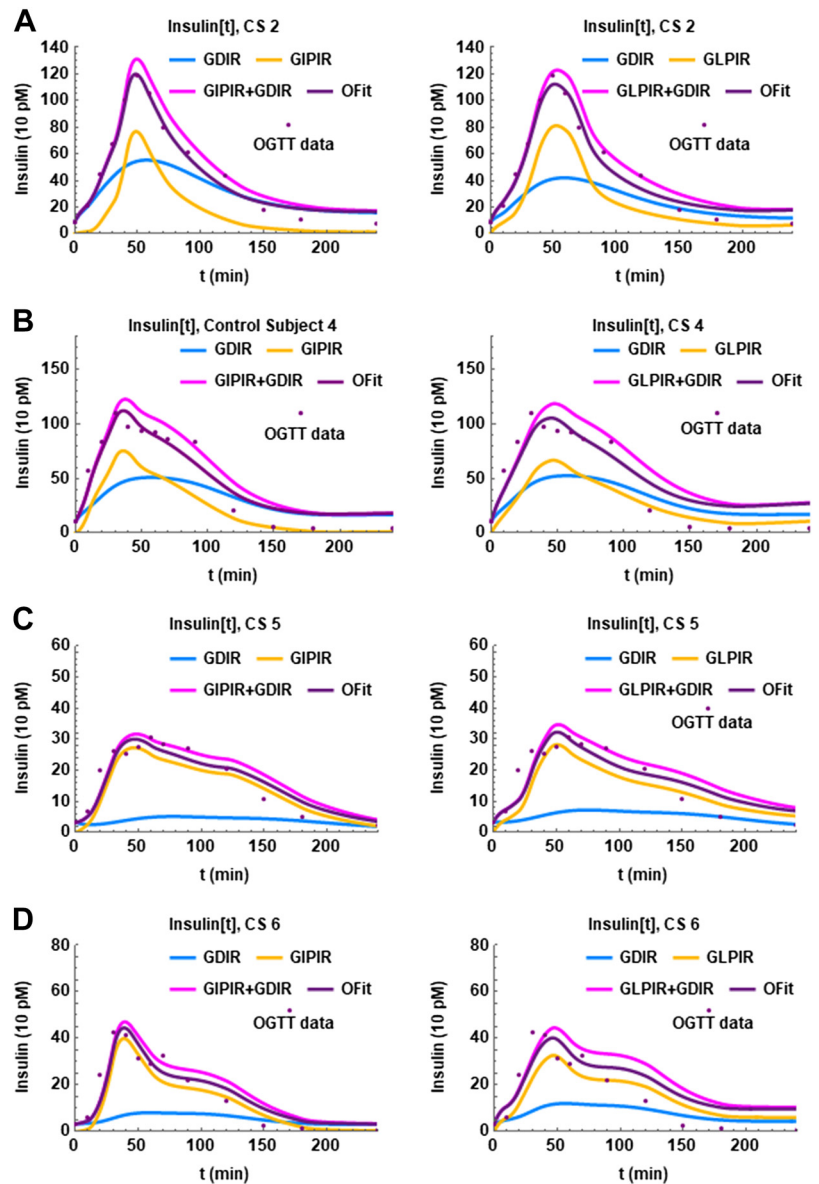


Figure 7. A plot of the insulin profiles obtained in step 1, 2, and 3 described in *Estimation of the Incretin Effect from OGTT* is presented below for four representative control subjects (75 g oral glucose tolerance test, OGTT; A–D). The area under the curve of these insulin profiles were used to calculate the incretin effect. In each panel, the results obtained with glucose-dependent insulinotropic polypeptide (GIP) or glucagon-like peptide-1 (GLP-1) fit for each subject is presented. The blue curve is the glucose-dependent insulin response (GDIR; step 2 simulation). The yellow curve is the incretin-dependent insulin response (GIPIR or GLPIR; step 3 simulation). The purple curve is the original fit (OFit; step 1). The magenta curve is the sum of the GDIR and GIPIR or GLPIR (blue and yellow curves).

apparent. The Pearson correlation coefficients and associated *P* values in parenthesis for the GIP fit, the GLP-1 fit, and average calculated values with the experimental values (*method 1*) were 0.71 (<0.006), 0.53 (<0.06), and 0.68 (<0.0099), respectively. The Spearman rank test correlation coefficients were 0.74 (<0.003), 0.44 (<0.12), and 0.67 (<0.012), respectively. The Pearson correlation coefficients and associated *P* values in parenthesis for the GIP fit, the GLP-1 fit, and average calculated values with the experimental values (*method 2*) were 0.82 (<0.00056), 0.67 (<0.011), and 0.81 (<0.00064) respectively. The Spearman rank test correlation coefficients and associated *P* values were 0.80 (<0.0001), 0.54 (<0.057), and 0.74 (<0.0041), respectively. The individuals with negative experimental incretin effect were not included in the statistics.

Trends in parameters.

The parameters determined by fitting the model to the data (GLP-1 fit, 7 CS and 8 T2D individuals) were plotted against

the three markers of T2D: HbA1c, fasting glucose (FG), and 2 h plasma glucose (2 h PG) levels post-OGTT in Fig. 10, A–F to see how they correlate.

In Fig. 10A, the glucagon suppression constant, k_1 , showed a linear correlation with all three markers of diabetes. The Spearman rank test correlation coefficient and associated *P* values in parenthesis were: –0.54 (<0.036), –0.74 (<0.0015), and –0.70 (<0.003), respectively. This behavior was also observed when modeling IIGI.

The measure of insulin sensitivity, a_1 , was found to separate into two quadrants as seen in Fig. 10B. Two CS, 2 and 4, had lower insulin sensitivity, in the range estimated in individuals with T2D. The insulin sensitivity parameter was uniformly lower in individuals with T2D who showed elevated HbA1c, FG, and 2 h PG but there was no linear correlation. The exact functional dependence of these markers on insulin sensitivity is not known but as seen in Fig. 10B, it appears to be a simple inverse (hyperbolic) relationship. To test this, the inverse of the insulin sensitivity parameter, $1/a_1$ (insulin

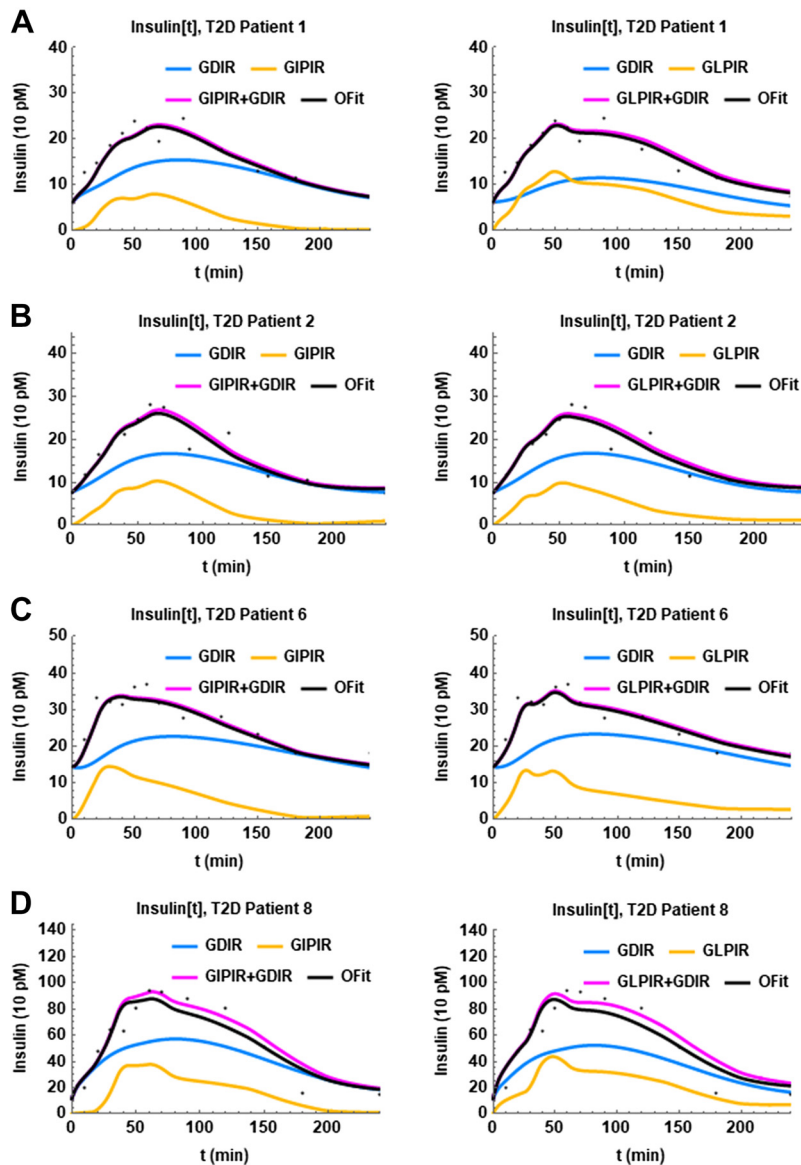


Figure 8. A plot of the insulin profiles obtained in step 1, 2, and 3 described in *Estimation of the Incretin Effect from OGTT* is presented below for four individuals with type 2 diabetes (T2D) (75 g oral glucose tolerance test, OGTT; A–D). The area under the curve of these insulin profiles were used to calculate the incretin effect. In each panel, the results obtained with glucose-dependent insulinotropic polypeptide (GIP) or glucagon-like peptide-1 (GLP-1) fit for each subject is presented. The blue curve is the glucose-dependent insulin response (GDIR; step 2 simulation). The yellow curve is the incretin-dependent insulin response (GIPIR or GLPIR; step 3 simulation). The purple curve is the original fit (OFit; step 1). The magenta curve is the sum of the GDIR and GIPIR or GLPIR (blue and yellow curves).

resistance), was plotted against HbA1c, FG, and 2 h PG as shown in Fig. 11A. A linear correlation is observed with Spearman rank test correlation coefficients and *P* values in parentheses of 0.62 (<0.013), 0.78 (<0.0005), 0.72 (<0.002), respectively. The slope and intercept in parentheses were 3,276 (–13,788), 149 (–11,372), and 34 (–225), respectively.

The glucagon action parameter, a_2 , was similar in CS and individuals with T2D as seen in Fig. 10C, and does not correlate with elevated HbA1c, FG, and 2 h PG. The glucose-dependent insulin secretion, γ_1 , is similar in CS and individuals with T2D as seen in Fig. 10D, and does not correlate with elevated HbA1c, FG, and 2 h PG.

The glucagon secretion parameter γ_2 separates into two quadrants in Fig. 10E and is higher in individuals with T2D. Higher values of γ_2 observed in individuals with T2D correlate with higher HbA1c, FG, and 2 h PG though again a simple linear relationship is not observed.

The parameter that is a measure of incretin-dependent insulin secretion γ_3 is seen to separate into two quadrants,

especially with respect to FG and 2 h PG in Fig. 10F. All the individuals with T2D have low incretin-dependent insulin secretion relative to CS except T2D 8. Here again, though the exact functional relationship of γ_3 with the markers HbA1c, FG, and 2 h PG is not known it appears to be a simple hyperbolic relationship. In Fig. 11B, the inverse of γ_3 is plotted against HbA1c, FG, and 2 h PG. A linear relationship is observed. The Spearman rank test correlation coefficients and *P* values are 0.81 (<0.0005), 0.71 (<0.0038), and 0.78 (<0.0008), respectively. The slope and intercept of the best-fit lines are 296 (–1418), 823 (–10), and 3.27 (–196).

Effects of Glucose Dose in OGTT

The OGTT experiments were carried out on the same individuals for three different doses of ingested glucose, 25 g, 75 g, and 125 g. The results for the 25 g and 125 g OGTT experiments are presented in Supplemental Section S1. An in-depth comparison is then made between the results from the

Table 4. The calculated and experimentally determined incretin effect % in CS and T2D individuals are presented

Patient No.	Incretin Effect %							
	CS 75 g OGTT				T2D 75 g OGTT			
	GIP	GLP-1	Average	Experimental I/II	GIP	GLP-1	Average	Experimental I/II
1	53	46	50	55 (63)	21	44	33	35 (43)
2	35	49	42	40 (44)	23	25	24	-32 (-36)
3					0	0	0	-8 (-25)
4	42	46	44	27 (45)	40	52	46	33 (37)
5	78	71	75	57 (84)	28	58	43	4 (11)
6	69	66	68	41 (83)	22	24	23	-5 (-11)
7	87	53	70	62 (75)	28	47	38	17 (40)
8	49	60	55	67 (86)	28	36	32	45 (58)

Calculated estimates from the GLP-1 and GIP fits and the average of the two estimates are in columns 2–4 and 6–8. The experimentally estimated values are in columns 5 and 9. GIP, glucose-dependent insulintropic polypeptide; GLP-1, glucagon-like peptide-1; OGTT, oral glucose tolerance test.

three data sets. The original study (17) was intended to estimate the influence, if any, of the glucose dose on the incretin effect. Here, we show through modeling and fitting the data, the effect it has on the incretin effect as well as the parameters related to glucose-insulin-glucagon dynamics.

Comparison of the incretin effect.

The incretin effect calculated as in *Estimation of the Incretin Effect from OGTT* from the 25 g, 75 g, and 125 g OGTT data is compared with to see if there is a glucose dose dependence. In Table 6, the mean estimated values of the incretin effect from the GIP and GLP-1 fit and the Mann-Whitney *U* test and Wilcoxon rank test *P* values are presented. The comparison was done on paired individuals as fits were not obtained for all the subjects for the different glucose doses.

When comparing the 25 g with 75 g and 125 g results in CS, the GIP incretin effect is not significantly different. The GLP-1 incretin effect is however significantly different between 25 g and 75 g. There are too few individuals to compare the GLP-1 effect between 25 g and 125 g. The GIP and GLP-1 incretin effect was not significantly different between the 75 g and 125 g OGTT.

Comparing the 25 g and 75 g results in T2D, the GIP and GLP-1 incretin effect are both not significantly different. Between 25 g and 125 g, the GIP effect is not significantly different. There are too few individuals to compare the GLP-1 effect between 25 g and 125 g. The GIP incretin effect is not

significantly different between 75 g and 125 g, but the *P* values are borderline.

Comparing the incretin effect in CS with T2D, the incretin effect is less in T2D and significantly different between CS 25 g and T2D 25 g as well as CS 75 g and T2D 75 g. The GIP-based incretin effect is found to be significantly different between CS 125 g and T2D 125 g, based on the Students *t* test. The GLP-1-based incretin effect is not significantly different between CS 125 g and T2D 125 g.

Comparison of the parameters.

The parameters estimated from fitting the model to the 25 g, 75 g, and 125 g data are presented in Table 7. Parameters that are significantly different between the groups are listed in the table. The Wilcoxon signed-rank test at the 5% significance level was used to make the comparisons within a group and the Mann-Whitney *U* test at the 5% significance level was used for between-group comparisons.

In the CS group, the glucagon action parameter, a_2 , showed a significant dose dependence. This parameter is a measure of glucagon-induced hepatic glucose production and the results show that as the amount of glucose ingested increases, the greater the suppression of hepatic glucose production (HGP). The value of the parameter drops from an average value of 0.43 in the 25 g OGTT estimate to 0.10 in the 125 g OGTT estimate. No other parameters were found to be significantly different.

Table 5. Comparison of the calculated incretin effect within groups and between groups (7 CS and 8 T2D) is presented in rows 1–5

No.	Comparison of Incretin Effect	<i>P</i> Value
1	CS 75 g OGTT (GIP) vs. CS 75 g OGTT (GLP)	0.93
2	T2D 75 g OGTT (GIP) vs. T2D 75 g OGTT (GLP)	0.022
3	CS 75 g OGTT (GIP) vs. T2D 75 g OGTT (GIP)	0.0014
4	CS 75 g OGTT (GLP) vs. T2D 75 g OGTT (GLP)	0.024
5	CS 75 g OGTT (Ave) vs. T2D 75 g OGTT (Ave)	0.032
6	CS 75 g OGTT (GIP) vs. experimental incretin effect <i>method 1 (method 2)</i>	0.52 (0.30)
7	CS 75 g OGTT (GLP) vs. experimental incretin effect	0.60 (0.25)
8	CS 75 g OGTT (Ave) vs. experimental incretin effect	0.48 (0.16)
9	T2D 75 g OGTT (GIP) vs. experimental incretin effect	0.83 (0.45)
10	T2D 75 g OGTT (GLP) vs. experimental incretin effect	0.036 (0.25)
11	T2D 75 g OGTT (Ave) vs. experimental incretin effect	0.34 (0.92)

A comparison of the calculated and experimentally determined incretin effects in seven controls (CS) and five individuals with type 2 diabetes (T2D) (only 5 T2D had positive experimental incretin effect) are presented in rows 6–8 and 9–11. The *P* values from the Mann-Whitney *U* test are presented. CS, control; GIP, glucose-dependent insulintropic polypeptide; GLP-1, glucagon-like peptide-1; OGTT, oral glucose tolerance test; T2D, type 2 diabetes.

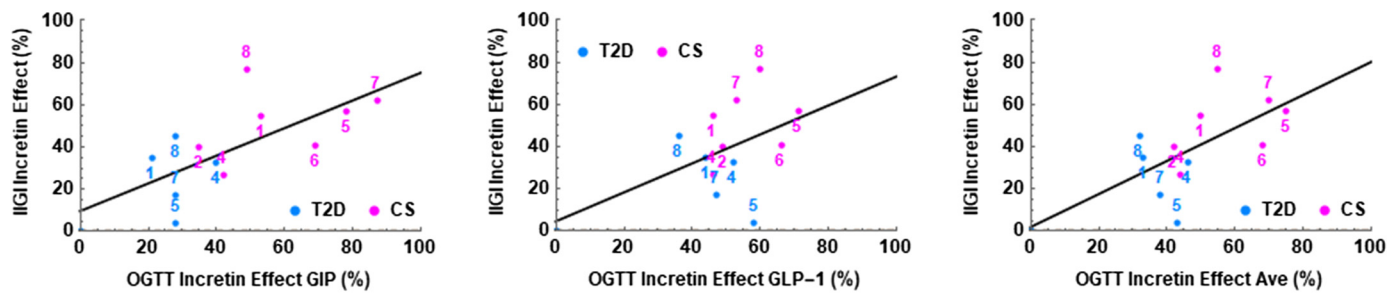
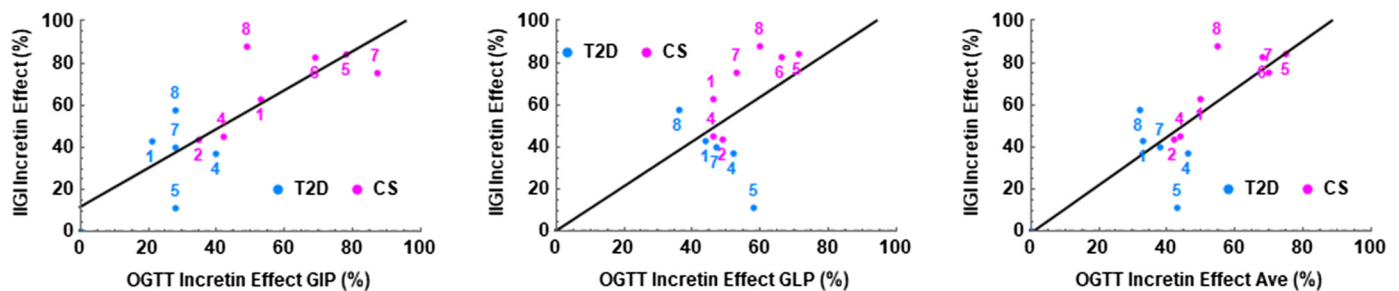
A Exp. method 1**B Exp. method 2**

Figure 9. Correlation between calculated incretin effect (IE) (x axis) and experimentally determined incretin effect (y axis). The experimental incretin effect was estimated from the areas under the curve of insulin in the oral glucose tolerance test (OGTT) and the corresponding isoglycemic intravenous glucose infusion (IIGI) using *methods 1 and 2*. The incretin effect calculated from the glucose-dependent insulinotropic polypeptide (GLP)-, glucagon-like peptide-1 (GLP-1)-fit, and the average values are compared with the experimental values. *A*: calculated IE (75 g OGTT) vs. experimental IE, *method 1* (75 g OGTT and corresponding IIGI). The Spearman rank test correlation coefficients were ($n = 12$): 0.74 (<0.003), 0.44 (<0.12), and 0.67 (<0.012), respectively. *B*: calculated (75 g OGTT) and experimental IE, *method 2* (75 g OGTT and corresponding IIGI). The Spearman rank test correlation coefficients and associated P values were ($n = 12$, 5 CS and 7 T2D): 0.80 (<0.0001), 54 (<0.057), and 0.74 (<0.0041), respectively.

In the T2D cohort, comparing the parameter estimates from the 25 g and 75 g OGTT, the insulin sensitivity parameter, a_1 , was found to be significantly different with a modest elevation in the average value of the parameter estimate (0.00025 vs. 0.00011). No other parameters were significantly different. No significant differences were found when comparing the parameter estimates between 75 g and 125 g OGTT as well as between 25 g and 125 g OGTT.

The estimated glucagon suppression parameter, k_1 , and the measure of incretin effect, γ_3 , were found to be significantly different between CS and T2D in all three experiments, 25 g, 75 g, and 125 g. The estimated insulin sensitivity parameter, a_1 , was significantly different between CS and T2D in the 75 g and 125 g experiments. In the 25 g experiment, a_1 was not significantly different between CS and T2D though the mean a_1 was higher for CS.

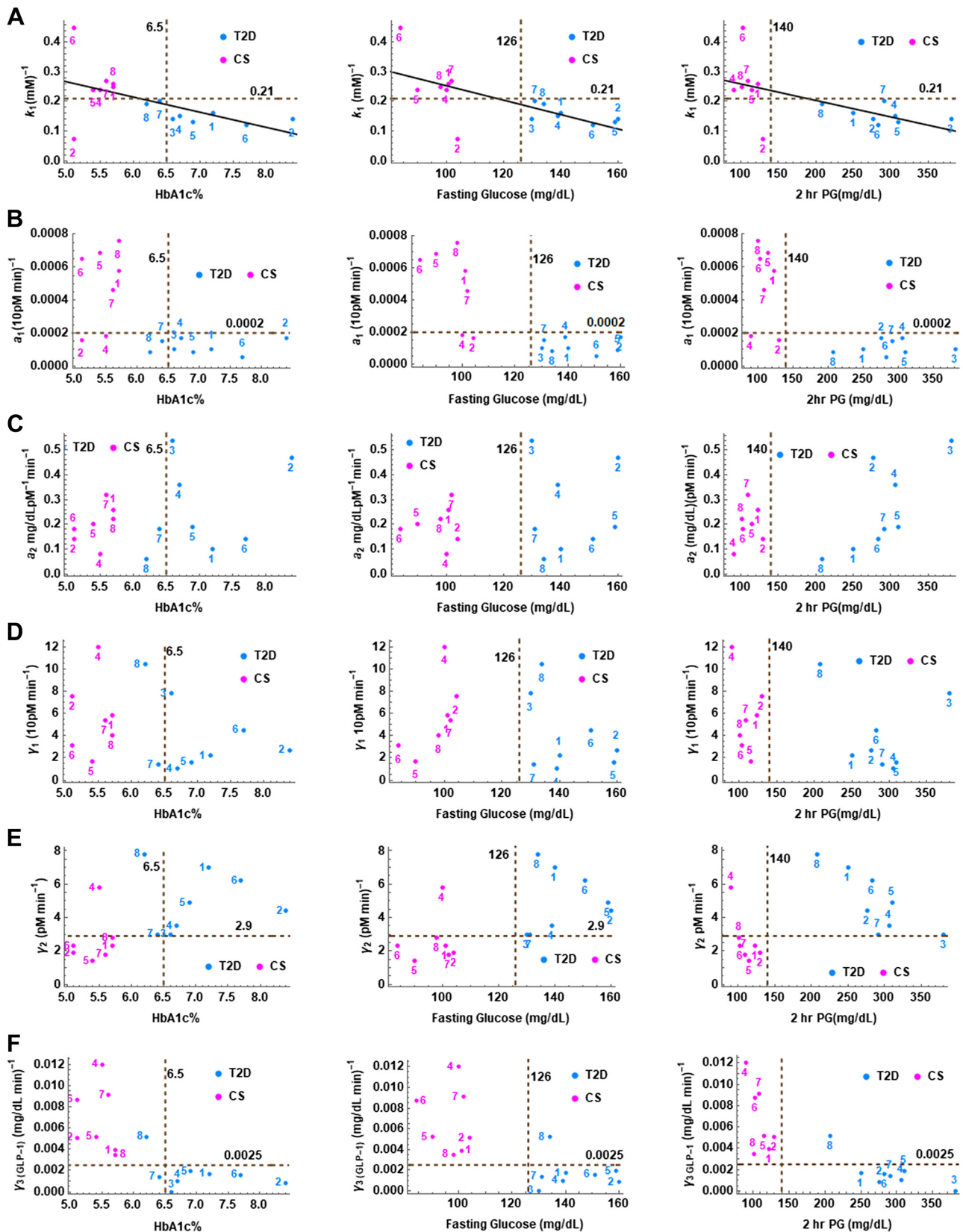
DISCUSSION

The primary aims of this paper were to quantify the role of α -cell dysfunction in T2D as well as directly estimate the incretin effect and its impairment in T2D by modeling OGTT experiments. The model developed is an extension of the minimal model used to describe the IIGI experiment in a previously published paper by us (49). The coupled model can be used to estimate parameters related to α -cell function, i.e., glucagon secretion and suppression, as well as glucagon action. In addition, glucose- and incretin-dependent insulin secretion that describes β -cell function and insulin sensitivity

can also be estimated by fitting data from OGTT experiments. The parameters estimated were then used to calculate the incretin effect by carrying out additional simulations in the presence and absence of the incretin or glucose-dependent contribution to insulin secretion.

The minimal model was fitted to OGTT data from CS and individuals with T2D. A new parameterized function, the skew normal distribution, was used to model the glucose arrival from the gut. The glucose arrival rate tends to be bimodal in CS and is more rapid than in the individuals with T2D. One reason for the slow arrival in individuals with T2D is likely due to the unfavorable concentration gradient for the movement of glucose into the capillaries from the interstitial space near the intestinal epithelium. This is because basal glucose levels are very high in T2D relative to CS. Gastric emptying was not found to be different between CS and T2D (17) and is not the reason for the differences in profiles.

The estimation of parameters related to glucagon suppression, secretion, and action by fitting data from the 75 g OGTT experiment showed that the glucagon suppression constant, k_1 , was significantly lower in individuals with T2D relative to CS and likely contributes to its pathophysiology. This defect in glucagon suppression has been previously inferred by researchers based on various experiments (79–83). Chen et al. (84) developed a quantitative rubric based on glucagon and insulin ratios, though there is evidence in the literature that insulin influences glucagon secretion only at very high levels (85, 86). Here, we have quantified it from a fully



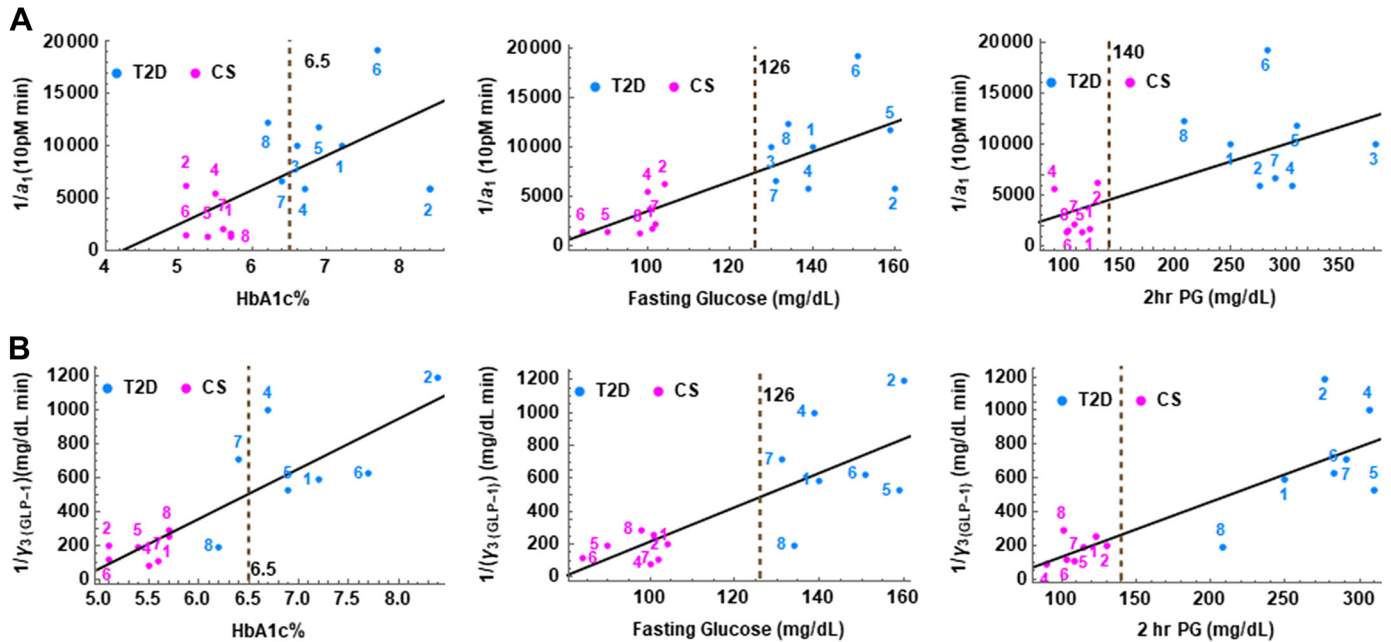


Figure 11. In A, a plot of the inverse of the insulin sensitivity parameter, $1/a_1$, vs. HbA1c, fasting glucose (FG), and 2 h plasma glucose (2 h PG) is presented. A linear correlation is observed with Spearman rank test correlation coefficients and P values in parentheses of ($n = 15$, 7 CS and 8 T2D): 0.62 (<0.013), 0.78 (<0.0005), and 0.72 (<0.002), respectively. The slope and intercept in parentheses are 3276 (–13788), 149 (–11372), and 34 (–225), respectively. In B, the inverse of the glucagon-like peptide-1 (GLP-1)-dependent insulin secretion parameter, $1/\gamma_3$ vs. HbA1c, FG, and 2 h PG is plotted. A linear correlation is observed in all three cases. The Spearman rank test correlation coefficients and P values are ($n = 15$): 0.81 (<0.0005), 0.71 (<0.0038), and 0.78 (<0.0008), respectively. The slope and intercept of the best fit lines are 296 (–1418), 823 (–10), and 3.27 (–196). CS, control; T2D, type 2 diabetes.

coupled model of OGTT with glucagon suppression controlled by glucose, for which there is significant evidence, particularly in the early postprandial phase where paracrine influence is expected to be low (57, 87). The parameter that is a measure of glucagon secretion, γ_2 , was also found to be higher in individuals with T2D relative to CS in the 75 g OGTT experiment. This is in line with the experimental studies by Unger et al. (23, 24) where they showed that baseline glucagon levels were high for the level of prevailing glycemia in T2D though not necessarily higher than in CS. A time delay was observed in the suppression of glucagon secretion in T2D relative to CS, though the etiology is uncertain.

The insulin sensitivity parameter, a_1 , was found to be higher in CS relative to T2D when estimated from the 75 g OGTT. This is in agreement with the results in the literature (88) regarding loss of insulin sensitivity in T2D. The measure of glucose-dependent insulin secretion, γ_1 , was not found to be significantly different between CS and T2D. One reason might be that glucose-stimulated second-phase insulin secretion is known to be relatively preserved in early T2D unlike first-phase insulin secretion which is completely lost (89). No significant difference is seen in second-phase insulin secretion between controls and early

T2D in bolus intravenous glucose infusion experiments (89, 90), for example, though this has been attributed to the elevated ambient glucose levels in T2D. Similarly in the companion IIGI experiments to the OGTT in the cohort studied in our paper (17) and in Nauck et al. (91), average insulin secretion was higher in T2D than in CS. In hyperglycemic clamp experiments where glucose levels are clamped at the same level in T2D and CS, differences in second-phase insulin secretion are observed but found to be relatively preserved in early T2D (89, 92). There is no signature of first-phase insulin secretion in response to OGTT and likely contributes less than in bolus insulin experiments (~15%) to overall secretion (93–95). Second, in early T2D, some level of compensatory insulin secretion might persist though not sufficient for the level of insulin resistance. Thus overall, the glucose-stimulated insulin secretion constants in early T2D are found to be similar to CS in our study. An experiment that would validate this observation (which was not done on our cohort) would be a glucose infusion experiment as described in the paper by Perley and Kipnis (96). In the paper, the authors studied insulin secretion in infusion-maintained plasma glucose profiles characteristic of T2D in both normal and subjects

Figure 10. Trends in parameters k_1 , a_1 , a_2 with respect to HbA1c, fasting glucose (FG), and 2 h plasma glucose (2 h PG) are shown in A–C. In A, the glucagon suppression, k_1 , constant shows linear correlation with all three markers of type 2 diabetes (T2D). The Spearman rank test correlation coefficient and associated P values in parenthesis were ($n = 15$, 7 CS and 8 T2D), –0.54 (<0.036), –0.74 (<0.0015), and –0.70 (<0.003), respectively. In B, the insulin sensitivity parameter, a_1 , separates into two quadrants except for two outliers. The glucagon action parameter, a_2 , does not show any trends with respect to the markers of T2D. F: trends in parameters γ_1 , γ_2 , γ_3 with respect to HbA1c, FG, and 2 h PG are shown in D–F. In D, the parameter γ_1 , a measure of glucose-dependent insulin secretion does not show any correlation with HbA1c, FG, and 2 h PG. The glucagon secretion parameter, γ_2 , separates into two quadrants and is higher in individuals with T2D. The measure of glucagon-like peptide-1 (GLP-1)-dependent insulin secretion, γ_3 , is seen to separate into two quadrants and is much lower for individuals with T2D.

Table 6. Comparison of the calculated incretin effect in CS and T2D from the three different OGTT experiments, 25 g, 75 g, and 125 g

Incretin Effect Comparison	Mean Values GIP (GLP-1) %	Mann-Whitney U Test; Signed Rank Test P Value GIP (GLP-1)
CS 25 g vs. CS 75 g	65 (80); 59 (55)	0.56 (0.012); 0.55 (0.036)
CS 25 g vs. CS 125 g	73 (—); 64 (75)	0.91 (—); 1.
CS 75 g vs. CS 125 g	59 (53); 64 (75)	0.30 (0.34); 42 (0.50)
T2D 25 g vs. T2D 75 g	27 (50); 23 (37)	0.83 (0.20); 0.55 (0.21)
T2D 25 g vs. T2D 125 g	32 (—); 47 (60)	0.25 (—); 0.42 (—)
T2D 75 g vs. T2D 125 g	27 (36); 47 (60)	0.06 (0.075); 0.10 (0.17)
CS 25 g vs. T2D 25 g	65 (80); 27 (50)	0.003 (0.012)
CS 75 g vs. T2D 75 g	59 (56); 23 (36)	0.0014 (0.024)
CS 125 g vs. T2D 125 g	64 (75); 47 (60)	0.06 (0.91) TTest 0.04 (0.78)

Mean values are presented in column 2. In the first 6 rows within-group comparisons are presented. Mean values for paired individuals are presented as only five individuals could be fit in the 125 g oral glucose tolerance test (OGTT) study. Mann-Whitney U test and Wilcoxon signed-rank test P values are presented in column 3. 125 g comparisons: ($n = 5$, CS and T2D). 25 g vs. 75 g comparisons ($n = 6$, CS GLP-1; $n = 7$, CS GIP; $n = 7$ T2D GLP-1, $n = 8$ T2D GIP). CS, control; GIP, glucose-dependent insulinotropic polypeptide; GLP-1, glucagon-like peptide-1; T2D, type 2 diabetes.

with T2D. They found that normal weight individuals without diabetes had similar insulin secretory profiles to obese diabetic individuals. Similarly, in infusion-maintained normal plasma glucose profiles, the obese individuals with diabetes had higher insulin secretion levels relative to normal weight individuals without diabetes (Figs. 4 and 5 in the reference). In weight-matched subjects, the results were different. This indicates that insulin secretion is intricately linked with insulin sensitivity and compensation making it difficult to generalize the extent of reduction in glucose-stimulated insulin secretion, particularly in early T2D relative to normal individuals. Depending on the cohort studied, one might get differing results. The cohort in our paper is a mix of overweight and obese subjects based on body mass index (BMI) in both T2D and CS groups.

The parameter, γ_3 , that is a measure of the incretin-dependent insulin secretion was found to be significantly lower in individuals with T2D relative to CS. This is in agreement with the results by Bagger et al. (17). Although the parameter, γ_3 , is a measure of incretin-dependent insulin secretion, it

does not give us the relative contribution of the incretins to the overall insulin secretion vis-à-vis glucose-dependent insulin secretion. To estimate the incretin effect, the percentage contribution of the incretins to overall insulin secretion post glucose challenge, directly from the OGTT data, the methodology of *Estimation of the Incretin Effect from OGTT* was adopted. The model yields either the GLP-1 or GIP only incretin effect. A comparison was then made between the experimental (17) and GLP-1 or GIP only incretin effect as well as an average of the two effects estimated from the 75 g OGTT. Although the incretin effect determined experimentally was positive in the case of the CS, some of the individuals with T2D showed a negative incretin effect. No such negative incretin effect was observed in the incretin effect determined from modeling and simulations (steps 1–3 and Eq. 8). The calculated incretin effect was found to correlate well with the experimentally determined incretin effect when subjects, both CS and individuals with T2D, with positive incretin effect were compared. The incretin effect was found to be significantly lower in individuals with T2D relative to CS. One patient with T2D was found to have no incretin effect. These results are in line with those observed experimentally (97, 98). Overall, this computational method for assessing incretin effects from the OGTT data appears to be a promising alternative to carrying out two experiments namely the OGTT followed by IIGI, on different days to obtain an estimate of the incretin effect. Second, as the incretin effect in T2D is difficult to assess accurately via experiment, this might be a better way of discriminating between individuals with T2D. Although Dalla Man et al. (43) estimate the incretin effect from one mixed meal tolerance test, they do so by making the assumption that the incretin-induced insulin secretion is proportional to the glucose-stimulated insulin secretion. No such assumption is made in our model. On the other hand, the glucose dependence of the incretin-induced insulin secretion is well documented and is implemented in our model (62). One area of improvement of the current model would be to integrate it with a model for incretin secretion from the gut and subsequent dynamics as described by Røge et al. (99).

The importance of incretins in glucose homeostasis is apparent from the efficacy of GLP-1 receptor agonists in treating individuals with T2D. To assess the extent to which

Table 7. A comparison of the parameters a_1 , a_2 , k_1 , k_2 , γ_1 , γ_2 , and γ_3 estimated from the 25 g, 75 g, and 125 g OGTT data is presented

Comparison	Parameters That Are Significantly Different ($n = 5$, Mean Values; Wilcoxon Signed-Rank Test P Value)
CS 25 g vs. CS 75 g (GIP fit paired)	a_2 (0.43, 0.26; <0.036)
CS 25 g vs. CS 125 g (GIP fit paired)	a_2 (0.45, 0.10; <0.036)
CS 75 g vs. CS 125 g (GIP fit paired)	a_2 (0.20, 0.10; <0.036)
T2D 25 g vs. T2D 75 g (GIP fit paired)	a_1 (0.00025, 0.00011; <0.031; <0.042)
T2D 75 g vs. T2D 125 g (GIP fit paired)	
T2D 25 g vs. T2D 125 g (GIP fit paired)	
Parameters That Are Significantly Different (Mean Values; Mann-Whitney U Test P Value)	
($n = 7$) CS 25 g vs. ($n = 8$) T2D 25 g (GIP fit)	k_1 (0.35, 0.16; <0.01) γ_3 (0.0043, 0.00094; <0.0045)
($n = 7$) CS 75 g vs. ($n = 8$) T2D 75 g (GIP fit)	a_1 , γ_3 , k_1 , γ_2 (ref. Table 1)
($n = 5$) CS 125 g vs. ($n = 5$) T2D 125 g (GIP fit)	$a_1 \times 10^{-4}$ (3.7, 1.2; <0.003), k_1 (0.24, 0.12; <0.036), γ_3 (0.0049, 0.0013; <0.021)

For each comparison, only the parameters that were determined to be significantly different by the Wilcoxon signed rank test or Mann-Whitney U test at the 5% significance level are shown in column 2 followed by the mean values of these parameters and the P values in parentheses. CS, control; GIP, glucose-dependent insulinotropic polypeptide; OGTT, oral glucose tolerance test; T2D, type 2 diabetes.

increasing GLP-1 concentrations to values comparable with weekly administration of the analog exenatide affects postprandial glucose levels, *Eqs. 1–3* were simulated with the parameters estimated for *T2D 1* and *8* from the 75 g OGTT at three different concentrations of GLP-1. In the paper on the pharmacokinetics of exenatide by Fineman et al. (100), weekly dosing of 2 mg extended release of exenatide results in a mean plasma level of ~ 250 pg/mL or 60 pM (Fig. 2 in Ref. 100). Assuming that the intrinsic efficacy of GLP-1 is comparable with exenatide, carrying out simulations with levels of GLP-1 similar to that of analog + endogenous GLP-1

production should give us an idea of how much postprandial glucose control can be achieved by administering the drug. The simulations were carried out under three different conditions: 1) at a constant concentration of 110 pM over the course of the simulation, 2) at a constant concentration of 150 pM, and 3) five times the GLP-1 concentrations as observed in the 75 g OGTT. The results are shown in *Figs. 12* and *13*. In the case of *patient 1*, there is a lowering of the glucose maximum by ~ 30 –50 mg/dL and in *T2D 8*, ~ 40 –60 mg/dL depending on the GLP-1 concentration used. In Ref. 101, the mean reduction in postprandial glucose with the weekly 2 mg dosage was -57 ± 13 mg/dL.

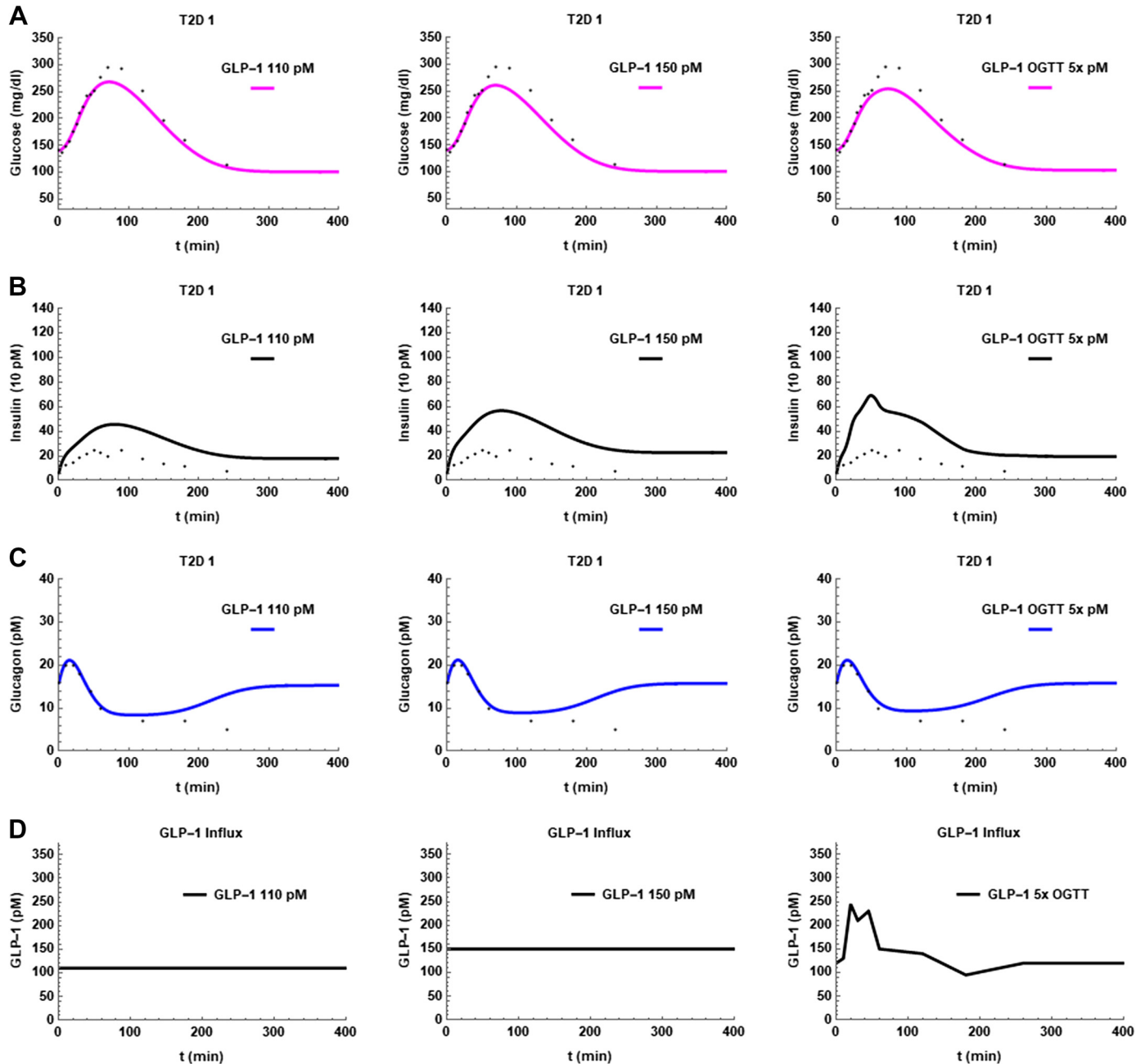


Figure 12. Simulations with three different concentrations of glucagon-like peptide-1 (GLP-1) in type 2 diabetes (*T2D*) *patient 1*, constant concentration of 110 pM, 150 pM, and a variable concentration of 5× the GLP-1 concentration from the 75 g oral glucose tolerance test (OGTT) data. The simulated glucose, insulin, and glucagon profiles are superimposed on the data from 75 g OGTT in rows A–C, respectively. The GLP-1 profile used in the simulations is presented in row D.

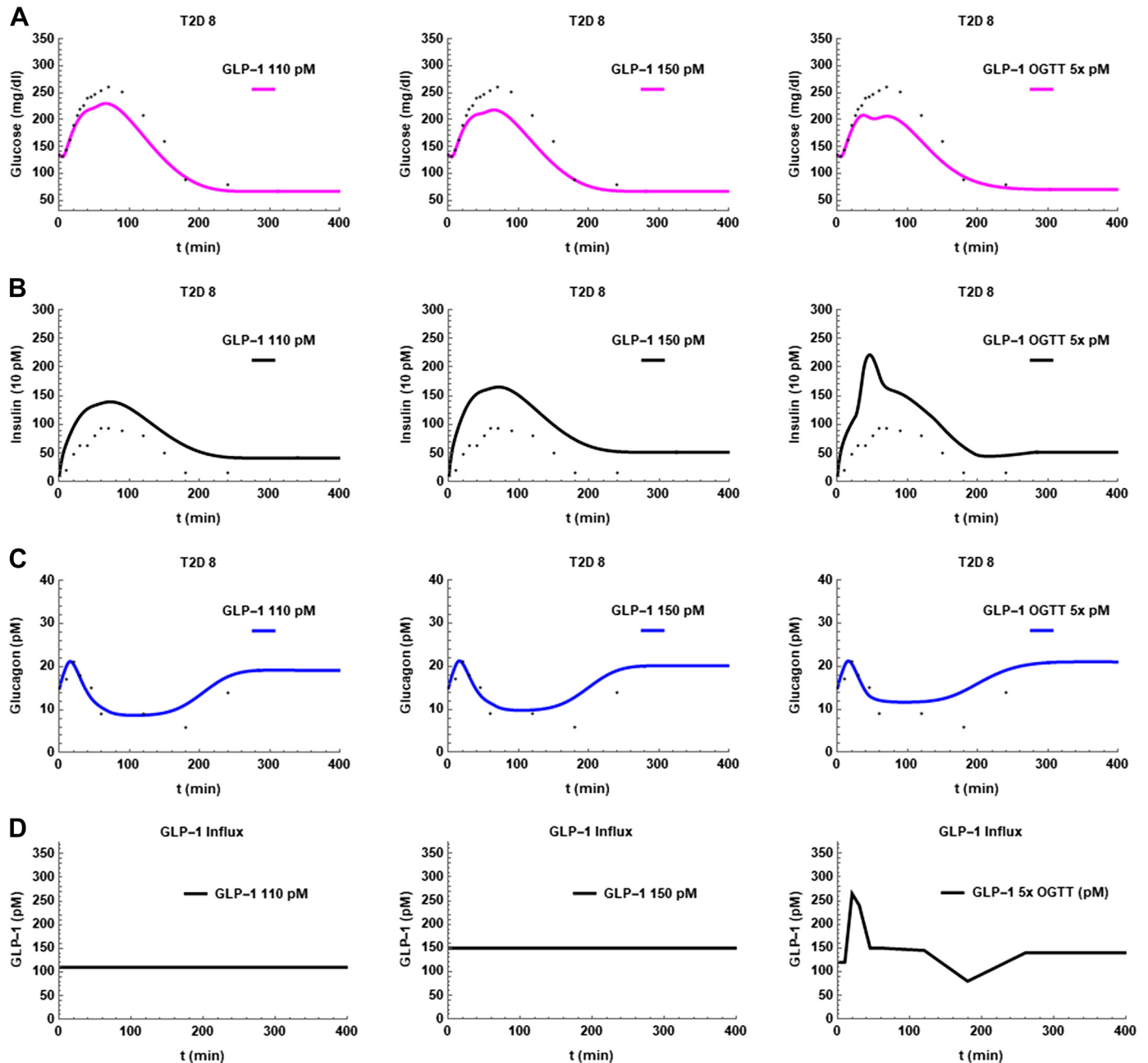


Figure 13. Simulations with three different concentrations of glucagon-like peptide-1 (GLP-1) in type 2 diabetes (T2D) *patient 8*, constant concentration of 110 pM, 150 pM, and a variable concentration of 5 $\times$ the GLP-1 concentration from the 75 g oral glucose tolerance test (OGTT) data. The simulated glucose, insulin, and glucagon profiles are superimposed on the data from 75 g OGTT in rows A–C, respectively. The GLP-1 profile used in the simulations is presented in row D.

Trends in the parameters estimated from fitting the model with respect to the markers of T2D, HbA1c, FG, and 2 h PG give us an idea of how impairment in glucagon secretion, suppression, action, insulin sensitivity, and glucose- and incretin-dependent insulin secretion affect postchallenge glucose excursions as well as glucose homeostatic set points. The main aim of this paper was to assess α -cell dysregulation or dysfunction as well as impairment in the incretin effect in T2D.

The glucagon suppression parameter shows negative linear correlation with HbA1c, FG, and 2 h PG. The correlation

coefficient was stronger with FG and 2 h PG than HbA1c. There is significant lowering of glucagon suppression in T2D, and it is likely to have an impact on both fasting and postprandial glucose levels as indicated by the correlation with these markers. In addition, glucagon secretion parameter, γ_2 , is elevated in T2D that correlates with higher HbA1c, FG, and 2 h PG but not in a linear manner like glucagon suppression.

The measure of incretin-dependent insulin secretion, γ_3 , is also significantly lower in T2D and correlates with higher levels of HbA1c, FG, and 2 h PG but not in a linear fashion. The plot of $1/\gamma_3$ versus HbA1c, FG, and 2 h PG on the other

hand showed a clear linear correlation. This indicates that postprandial glucose control via incretin-enhanced insulin secretion has a significant impact on all three markers of diabetes. The glucose-dependent insulin secretion parameter, γ_2 , on the other hand, does not show any relationship with the three markers for T2D and was also not significantly different between CS and T2D in this cohort.

The measure of insulin sensitivity, a_1 , separates into two quadrants and is much lower in T2D and correlates with higher HbA1c, FG, and 2 h PG. Two CS with a_1 values in the overlapping region between CS and T2D had high compensatory levels of insulin secretion, based on the calculated disposition index (DI) (Eq. 10) (71, 72, 74), which explains their inclusion in the CS group despite low insulin sensitivity. The box and whisker plot (101, 102) of the DI is shown in Fig. 14. The two CS with insulin sensitivity comparable with T2D individuals have a high enough insulin secretion index to move the DI up within 1.5 of the interquartile range of the CS group (the two DI values below the magenta box). The DI was also found to be significantly different between CS and T2D (Mann–Whitney U test: P value <0.0009). The plots of insulin resistance ($1/a_1$) versus the three markers show a linear relationship though there is some scatter, particularly in individuals with T2D. The weakest correlation was with HbA1c (0.64), whereas the correlation with FG and 2 h PG was comparable.

Modeling the dependence of the dynamics on glucose dose yielded significant information. The first significant result is the finding that the rate constant for hepatic glucose production, a_2 , shows a dose dependence in CS. As the glucose dose increases the rate constant is seen to decrease quite dramatically. This effect was not observed in the IIGI experiment [as seen in the Supplemental Material accompanying the IIGI paper (49)]. Although this effect may be attributed to the higher insulin secretion at higher glucose doses (86, 103, 104), there is also evidence in the literature that GLP-1 may have a direct suppressive effect on HGP (105–109). The second significant finding is that GIP is likely responsible for the incretin effect when smaller amounts of glucose are ingested. This has been noted in the literature (6,

110–112) and is attributed to the fact that GIP is secreted in the proximate region of the intestine, whereas GLP-1 is secreted in a more distal region with respect to the stomach.

T2D is a complex disease and a manifestation of dysfunction in multiple pathways involved in glucose homeostasis. Here we show that α -cell dysfunction can be quantified by mathematical modeling of the OGTT, and that impaired glucagon suppression and secretion also correlate with HbA1c, fasting glucose, and 2 h plasma glucose. We have also shown that the incretin effect can be estimated directly by fitting the coupled model to data from OGTT experiments and is significantly impaired in T2D. In addition, parameters related to insulin sensitivity and incretin secretion are also estimated simultaneously. One of the shortcomings of the model is the sensitivity to initial guess values in the nonlinear fitting procedure. It was found that Brent's method may be better than Levenberg–Marquardt in this respect. Newer methods of parameter estimation as in the paper by Bradley et al. (113) might help address problems due to noisy data and convergence to local minima and could be tested in the future. In their paper, a two-stage procedure is used. In the first step, parameters of a neural ODE (NODE) model are fitted using the measured data. In the second step, the parameters of the mechanistic model are found using the state and derivative estimates of the data-driven NODE model. The issue could also possibly be addressed by sampling the data more uniformly over the duration of the experiment. The current data set has too few points in the latter half of the experiment. Second the hormones, particularly, glucagon should be sampled more frequently as levels come back to baseline. Third, the experiment could be extended to 300 min to get a better picture of glucagon dynamics as the return to baseline values is very slow.

Future directions would include studying a larger cohort with this model and establishing cutoff values for the parameters that were found to be significantly different between CS and T2D. The linear correlations observed between insulin resistance, suppression of glucagon secretion, and inverse of the incretin effect parameter and fasting glucose, HbA1c, and 2 h PG could collectively be used to develop a rubric for identifying where a subject falls in the normal to T2D spectrum. In addition, using data from individuals before and after the use of incretin analog-based drugs would be useful in assessing the validity of the calculated incretin effect.

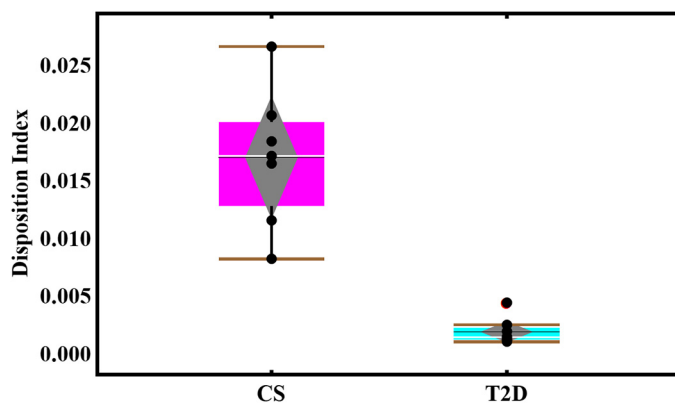


Figure 14. Box and whisker plot of the disposition index including outliers. There are no outliers in the control (CS) group while there is one in the type 2 diabetes (T2D) group. The disposition index (DI) values of individual subjects (black dots), median (white line), mean (black line), 95% confidence interval about the mean (gray diamond), outliers (red dot), and interquartile range (magenta-CS, cyan-T2D) are highlighted.

DATA AVAILABILITY

Data will be made available upon reasonable request.

SUPPLEMENTAL DATA

Supplemental Table S1, Supplemental Figs. S1 and S2, Supplemental Section S1, and Supplemental Material: <https://doi.org/10.7281/T1/UP8RWD>.

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DISCLAIMERS

The content of this manuscript is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

DISCLOSURES

V.S. and V.H. declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. J.J.H. is on the advisory committee of Novo Nordisk. F.K.K. has served on scientific advisory panels and/or been part of speaker's bureaus for, served as a consultant to, and/or received research support from Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, Carmot Therapeutics, Eli Lilly, Gubra, MedImmune, MSD/Merck, Mundipharma, Norgine, Novo Nordisk, Sanofi, ShouTi, Zealand Pharma and Zucara. T.V. has served on scientific advisory panels, been part of speaker's bureaus, and served as a consultant to, and/or received research support from Amgen, AstraZeneca, Boehringer Ingelheim, Eli Lilly, Gilead, GSK, Mundipharma, MSD/Merck, Novo Nordisk, Sanofi, and Sun Pharmaceuticals.

AUTHOR CONTRIBUTIONS

V.S. and J.I.B. conceived and designed research; J.I.B. performed experiments; V.S. and V.H. analyzed data; J.I.B., J.J.H., F.K.K., and T.V. interpreted results of experiments; V.S. and V.H. prepared figures; V.S. drafted manuscript; V.S., J.I.B., V.H., J.J.H., F.K.K., and T.V. edited and revised manuscript; V.S., J.I.B., V.H., J.J.H., F.K.K., and T.V. approved final version of manuscript.

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